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Recombinant polypeptides containing at least one immunogenic fragment and uses thereof

Abstract

Provided is a recombinant polypeptide containing at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein, and pharmaceutical compositions containing the same.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION (1) This application claims benefit to pending U.S. patent application Ser. No. 17/535,309, filed Nov. 24, 2021, which is a continuation application of International Patent Application No. PCT/US2021/040019, filed Jun. 30, 2021, which claims benefit to U.S. Provisional Patent Applications No. 63/046,426, filed Jun. 30, 2020, and 63/154,647, filed Feb. 26, 2021, all of which are each hereby incorporated by reference in their entireties.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

(1) Incorporated by reference in its entirety herein is a computer-readable nucleotide/amino acid sequence listing submitted concurrently herewith and identified as follows: One 848,455Byte ASCII (Text) file named “759990_ST25.txt,” created on Apr. 15, 2022.

BACKGROUND OF THE INVENTION

(2) The rapid evolution of new SARS-CoV-2 variants containing mutations that alter the amino acid sequence of the Spike protein resulting in altered function, and altered resistance to native immune defenses and to immune defenses elicited by currently marketed vaccines has led to a need for alternative platforms that allow for incorporation of new variant mutations in a robust vaccine.

BRIEF SUMMARY OF THE INVENTION

(3) The present invention relates to recombinant polypeptides that include at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and, optionally, an antibody Fc region. In some embodiments, the recombinant polypeptide includes more than one immunogenic fragment, e.g., two, three, four, five, or more immunogenic fragments. In some embodiments, the recombinant polypeptide includes one or more immunogenic fragments of Severe Acute Respiratory Syndrome Coronavirus (SARS-CoV) and/or Middle Eastern Respiratory Syndrome Coronavirus (MERS-CoV) and optionally an antibody Fc region, optionally in combination with one or more SARS-CoV-2 immunogenic fragments.

(4) The present invention further relates to pharmaceutical compositions, such as vaccines, that include the recombinant polypeptide. In some embodiments, the pharmaceutical composition includes an adjuvant.

(5) The present invention also relates to a method for preventing, inhibiting, reducing, eliminating, protecting, and/or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject, wherein the method includes administering to the subject at least one recombinant polypeptide of the invention or a pharmaceutical composition including the same.

(6) The present invention further relates to a method for inducing an immune response against a coronavirus in a subject, wherein the method includes administering to the subject at least one recombinant polypeptide of the invention or a pharmaceutical composition including the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a graph depicting Spike 51 protein IgG response in Rhesus macaques at certain time points (in weeks) after initial injection with an exemplary construct.

(2) FIG. 2A is an image showing CHO cell expression of CTD_short_a-Fc (LS2330) after 5 days.

Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining. M-Molecular weight markers. As can be seen in each of FIGS. 2A-2D, the constructs were resistant to proteolytic degradation during expression thus facilitating higher yields of intact, soluble protein

(3) FIG. 2B is an image showing CHO cell expression of CTD_long_a-Fc (LS3472) after 5 days. Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining. Protein loads per lane are indicated.

(4) FIG. 2C is an image showing CHO cell expression of CTD-deletion series after 5 days. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. Load volumes were twice as large for samples (left to right) CTD_vs_a-Fc to RBD_e-Fc compared to the samples on the left of the gel (CTD_long_b-Fc to CTD_short_h-Fc).

(5) FIG. 2D is an image showing CHO cell expression of CTD_short_i-Fc (LS2371) after 5 days. Protein was affinity purified using Protein A agarose and analyzed by reducing SDS-PAGE and detected by Coomassie R-250 staining.

(6) FIG. 3 is an image showing CHO cell expression of (CTD_short_d).sub.2-Fc (clone 1 through 4 correspond to strains LS2397 through 2400; SEQ ID NO: 161) and (CTD_short_i).sub.2-Fc (clone 1 through 4 correspond to strains LS2401 through 2404; SEQ ID NO: 163) after 4 or 7 days as indicated. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers.

(7) FIG. 4 is an image showing CHO cell expression of mutant (CTD_short_i).sub.2-Fc constructs wherein amino acid mutations corresponding to newly identified SARS-CoV-2 variants have been added to both CTD domains (D1 and D2) or second domain (D2) only of the CTD dimer. The mutants tested include (a) D1 and D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y (Strain 2435), (b) D2 mutations for 501.V2 variant/B.1.351, K417N, E484K, N501Y (Strains 2421), and (c) D2 mutations for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y (Strain 2423). Protein was affinity purified using Protein A agarose from equal volumes of 4-day transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers. Protein loads were 3.48 (strain 2435), 4.52 (strain 2421), and 3.14 μ g (strain 2423).

(8) FIGS. 5A, 5B, and 5C are graphs showing the results of neutralization assays using serum samples from animals P0101 (FIG. 5A) and P0102 (FIG. 5B) immunized with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc], which were the subject of analysis discussed in Example 1. FIG. 5C represents control samples including human serum from an individual prior to the emergence of SARS-CoV-2 (negative human serum sample) and the same serum spiked with a monoclonal antibody immunoreactive to the RBD portion of the Wuhan variant of the SARS-CoV-2 Spike protein. Infectivity of serum neutralized pseudotypes virus using 295T/ACE2 target cells was quantified by measuring NanoLuc luciferase activity (RLU) and graphed on the y-axis. Reciprocal serum dilution is shown on the x-axis.

(9) FIG. 6 is an image showing CHO cell expression of BA.X-1-monomer-Fc-BA.X-10-monomer-Fc after 4 days. Protein was affinity purified using Protein A agarose from equal volumes of transfected culture. Eluted protein was loaded based on volume, separated by reducing SDS-PAGE, and detected by Coomassie R-250 staining. M-Molecular weight markers. 'Transfect. Control' is an ACE2-Fc transfection, expression control.

DETAILED DESCRIPTION OF THE INVENTION

(10) Recombinant polypeptides of the invention can include any suitable immunogenic fragment or fragments of the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and optionally any suitable antibody Fc region. In some embodiments, an

immunogenic fragment comprises, consist of, or consist essentially of, the N-terminal domain of the S1 subunit, the C-terminal domain of the S1 subunit, or both. In certain embodiments, an immunogenic fragment can include the complete SARS-CoV-2 spike glycoprotein. Recombinant polypeptides of the invention include at least one immunogenic fragment, and can contain two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or more such immunogenic fragments.

(11) In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. Wild-type spike glycoproteins include those of any SARS-CoV-2 strain that has been isolated from a subject. Examples include Wuhan-Hu-1, VOC 202012/01/B.1.1.7 (Alpha or UK), VOC-202102/02 (B.1.1.7 with E484K) (UK), 501.V2/B.1.351 (South Africa), B.1.429/CAL.20C (California), B.1.525, and Lineage P.1 (Gamma or Brazil), B.1.427 (Epsilon), B.1.429 (Epsilon), B.1.617.1, B.1.617.2 (Delta), B.1.526 (Iota), B.1.617.3, B.1, A.2.5, C.36.3, B.1.1.318, B.1.351, B.1.621, B.1.525, P.1.1, P.2 (Zeta), B.1.623, R.1, B.1.1.7, B.1.351, B.1.351.3, B.1.351.3, BA.1, BA.1.1, BA.1.1.1, BA.1.1.2, BA.1.1.12, BA.1.1.13, BA.1.14, BA.1.15, BA.1.15.1, BA.1.16, BA.1.17.2, BA.1.18, BA.2, BA.2+, BA.2.1, BA.2.10.1, BA.2.2, BA.2.3.2, BA.2.4, BA.2.5, BA.2.6, BA.2.7, BA.2.8, BA.2.9, BA.2.10, BA.2.12, BA.3, BA.4, BA.5, XA, XB, XC, XD, XE, XF, XG, XH, XJ, XK, XL, XM, XN, XP, XQ, XR, XS, and XT. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV or MERS spike glycoprotein, or any portion thereof. Wild-type SARS-CoV and MERS spike glycoproteins include those of any SARS-CoV or MERS strain that has been isolated from a subject. Examples of such strains include SARS coronavirus Tor2 (GenBank accession number NC_004718.3) and MERS coronavirus (GenBank accession number NC_019843.3).

(12) In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a wild-type SARS-CoV or MERS spike glycoprotein, or any portion thereof.

(13) In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, K417, L452, S477, T478, E484, N501, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, G339, R346, 5371, 5373, 5375, T376, D405, R408, K417, N440, G446, L452, S477, T478, E484, Q493, G496, Q498, N501, Y505, T547, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence selected from SEQ ID NOs: 245-254, except for at one or more of the following positions of the amino acid sequence: L5, A67, H69, V70, D80, T95, G142, Y144, E154, F157, D253, G339, R346, 5371, S373, S375, T376, D405, R408, K417, N440, G446, L452, S477, T478, E484, Q493, G496, Q498, N501, Y505, T547, D614, Q677, P681, A701, T791, T859, F888, D950, and Q1071, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid

sequence selected from SEQ ID NOs: 245-254, except for at one or more of the following positions of the amino acid sequence: G339, R346, 5371, 5373, 5375, T376, D405, R408, K417, N440, G446, L452, 5477, T478, E484, Q493, G496, Q498, N501, and Y505, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). The mutations at these positions can be any suitable mutation, including conservative and non-conservative amino acid mutations. For instance, a conservative substitution can replace one aliphatic amino acid (i.e., Glycine, Alanine, Valine, Leucine, Methionine or Isoleucine) for another, one polar, uncharged R group amino acid (i.e., Serine, Cysteine, Threonine, Proline, Asparagine, or Methionine) for another, one positively charged R group amino acid (i.e., Histidine, Lysine, or Arginine) for another, one negatively charged R group amino acid (i.e., Aspartate or Glutamate) for another, and one non-polar, aromatic R group amino acid (i.e., Phenylalanine, Tyrosine, or Tryptophan) for another. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for one or more of the following amino acid substitutions and deletions: L5F, A67V, 69del, 70del, D80G, T95I, G142D, 144del, E154K, F157S, D253G, L452R, S477N, E484K, E484Q, K417N, K417T, T478K, N501Y, D614G, Q677H, T791I, P681H, P681R, A701V, F888L, T859N, D950H, D950N, and Q1071H, wherein the listed amino acid substitutions and deletions are relative to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In one embodiment, the one or more amino acid substitutions and/or deletions is L452R. In another embodiment, the one or more amino acid substitutions and/or deletions is E484K. In a further embodiment, the one or more protein substitutions and/or deletions are K417N, E484K, and N501Y. In yet another embodiment, the one or more substitutions are and/or deletions are K417T, E484K, and N501Y. In another embodiment, the one or more amino acid substitutions and/or deletions are N501Y, 69del, 70del, and P681H. In another embodiment, the one or more amino acid substitutions and/or deletions are K417T, L452R, E484K, and N501Y.

(14) In certain embodiments, one or more of the immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence comprising, consisting of, or consisting essentially of a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, wherein the amino acid sequence further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H to the extent the sequence does not already contain such residues. In some embodiments, one or more of the immunogenic fragments are identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof, except for one or more of the following amino acid substitutions and deletions: L5F, A67V, 69del, 70del, D80G, T95I, G142D, 144del, E154K, F157S, D253G, G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, N440K, G446S, L452R, S477N, E484A, E484K, E484Q, K417N, K417T, T478K, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, Q677H, T791I, P681H, P681R, A701V, F888L, T859N, D950H, D950N, and Q1071H, wherein the listed amino acid substitutions and deletions are relative to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In some embodiments, one or more of the immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence selected from SEQ ID NOs: 245-254, except for one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H, wherein the positions of the listed amino acid residues correspond to the wild-type amino acid sequence QHD43416 (ncbi.nlm.nih.gov/protein/QHD43416). In certain embodiments, one or more of the immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or

99%, sequence identity to an amino acid sequence selected from SEQ ID NOs: 245-254, wherein the amino acid sequence further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H, to the extent the sequence does not already contain such residues.

(15) In one embodiment, the one or more amino acid substitutions is G339D, R346K, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S373P, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484K, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, L452R, S477N, T478K, E484K, Q493R, and N501Y. In another embodiment, the one or more amino acid substitutions is G339D, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, and N501Y. In another embodiment, the one or more amino acid substitutions is G339D, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S371F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, R346K, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484K, Q493R, G496S, Q498R, N501Y, and Y505H. In another embodiment, the one or more amino acid substitutions is G339D, S373P, T376A, D405N, R408S, K417N, N440K, S477N, T478K, E484A, Q493R, Q498R, N501Y, and Y505H.

(16) The subject can be mammalian, including human, non-human primate, horse, pig, cattle, cat, dog, sheep, mink, rodent, hamster, or bat. Subjects can further include western lowland gorilla, northern white-cheeked gibbon, Sumatran orangutan, crab-eating macaque, drill, proboscis monkey, bonobo, chimpanzee, ugandan red colobus, red-shanked douc, golden snub-nosed monkey, green monkey, patas monkey, rhesus macaque, olive baboon, gelade, sooty mangabey, southern pig-tailed macaque, angola colobus, coquerel's sifaka, Gambian pouched rat, Chinese hamster, common gund, beluga whale, blue-eyed black lemur, indri, narwhal, narrow-ridged finless porpoise, harbour porpoise, minke whale, Antarctic minke whale, gray whale, spalax, white-tailed deer, reindeer, southern tamandua, Stephens's kangaroo rat, Pere David's deer, transcaucasian mole vole, long-finned pilot whale, Pacific white-sided dolphin, baiji, giant anteater, muskrat, killer whale, common bottlenose dolphin, aye-aye, fat-tailed dwarf lemur, thirteen-lined ground squirrel, yellow-bellied marmot, alpine marmot, golden hamster, sperm whale, daurian ground squirrel, gobi jerboa, barbry sheep, pronghorn, Nancy ma's night monkey, hirola, American bison, zebu, wild yak, cattle, water buffalo, white-eared titi, common marmoset, wild goat, goat, Panamanian white-faced capuchin, cat, Masai giraffe, Nilgiri tahr, Candian lynx, coquerel's giant mouse lemur, Siberian musk deer, sunda clouded leopard, clouded leopard, okapi, sheep, jaguar, leopard, Siberian tiger, Tibetan antelope, little pocket mouse, deer mouse, white-faced saki, cougar, black-capped squirrel monkey, tufted capuchin, arctic ground squirrel, *bos indicus* × *bos Taurus*, cheetah, mantled howler, Geoffroy's spider monkey, Damaraland mole-rat, naked mole-rat, hippopotamus, snowshoe hare, Dama gazelle, European rabbit, scimitar oryx, emperor tamarin, and alpaca.

(17) In some embodiments, one or more of the immunogenic fragments contain one or more mutations, such that the one or more immunogenic fragments are not identical to a wild-type SARS-CoV-2 spike glycoprotein, or any portion thereof. The one or more mutations can be any suitable mutation and/or deletion, such as those disclosed herein. For example, an immunogenic

fragment can contain sequences from two, three, four, five, six, seven, eight, nine, ten, or more strains, such that the resulting fragment is no longer identical to any of its parent strains. In this way, a single immunogenic fragment can present epitopes from multiple wild-type SARS-CoV-2 spike glycoproteins, or any portion thereof. This can lead to a more robust immune response and increased immune protection from a range of SARS-CoV-2 strains in a subject when a recombinant polypeptide of the invention containing one or more such immunogenic fragments, or a pharmaceutical composition containing the same, is administered to the subject.

(18) In some embodiments, the nucleic acid sequence encoding an immunogenic fragment includes one, two, three, four, five, six, seven, eight, nine, ten, twenty, thirty, forty, or more point mutations and/or deletions in comparison to the nucleic acid sequence encoding the corresponding fragment of a wild-type or mutant glycoprotein. In some embodiments, the amino acid sequence of an immunogenic fragment includes one, two, three, four, five, six, seven, eight, nine, ten, twenty, thirty, forty, or more substitutions and/or deletions in comparison to the amino acid sequence encoding corresponding fragment of a wild-type or mutant glycoprotein.

(19) In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight).sub.2), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended).sub.2), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD).sub.2), SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3), SEQ ID NO: 199 (SARS-2003, SARS_short_h), SEQ ID NO: 201 (SARS-2003, SARS_short_i), SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), SEQ ID NO: 215 (MERS_Lytic_g), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), and SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 185 ((CTD_short_i).sub.2). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ

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95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 266 (BA.X-2-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 267 (BA.X-3-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 268 (BA.X-4-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 269 (BA.X-5-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 270 (BA.X-6-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 271 (BA.X-7-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 272 (BA.X-8-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 273 (BA.X-9-dimer). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 274 (BA.X-10-dimer).

(20) In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274. In some embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), and SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3). In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence represented by SEQ ID NO: 185. In an embodiment, an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence represented by SEQ ID NO: 191. In further embodiments an immunogenic fragment comprises, consists of, or consists essentially of the amino acid sequence of SEQ ID NO: 185 ((CTD_short_i).sub.2). In further embodiments, an immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), and SEQ ID NO: 23 (CTD_short_a). In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 245-254 and 265-274. In some embodiments, the recombinant polypeptide comprises at least one immunogenic fragment that comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOS: 245-254. In some embodiments, the recombinant polypeptide

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three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments. In some embodiments, the recombinant polypeptide includes at least three immunogenic fragments. In some embodiments, the recombinant polypeptide includes at least three immunogenic fragments, of which at least one is a SARS-CoV-2 spike glycoprotein fragment as described herein, at least one is a SARS-CoV spike glycoprotein fragment as described herein, and at least one is a MERS-CoV spike glycoprotein fragment as described herein. In some embodiments, the recombinant polypeptide includes at least three immunogenic fragments, of which at least one is a SARS-CoV-2 spike glycoprotein fragment, wherein the SARS-CoV-2 fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight).sub.2), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended).sub.2), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD).sub.2), SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer); at least one is a SARS-CoV spike glycoprotein fragment, wherein the SARS-CoV fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 199 (SARS-2003, SARS_short_h), or SEQ ID NO: 201 (SARS-2003, SARS_short_i); and at least one is a MERS-CoV spike glycoprotein fragment, wherein the MERS-CoV fragment comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), and SEQ ID NO: 215 (MERS_Lytic_g).

(22) In some embodiments, the recombinant polypeptide includes at least two immunogenic fragments, wherein each of the at least two immunogenic fragments comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence independently selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO:

45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight).sub.2), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended).sub.2), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD).sub.2), SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In some embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence independently selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 23 (CTD_short_a), and SEQ ID NOs: 245-254 (BA.X-1-BA.X-10). In some embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 39 (CTD_short_i). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 23 (CTD_short_a),), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of the amino acid sequence of SEQ ID NO: 39 (CTD_short_i). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer). In further embodiments, each of the at least two immunogenic fragments comprises, consists of, or consists essentially of an amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10). The at least two immunogenic fragments can comprise, consist of, or consist essentially of, any suitable combination of amino acid sequences.

(23) In certain embodiments, each of the at least two immunogenic fragments comprise, consist of, or consist essentially of the same amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), e.g., the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 245. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of SEQ ID NO: 246. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 247. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 248. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 249. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 250. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 251. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 252. In another embodiment, the at least two immunogenic fragments each comprise, consist

of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 253. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 254. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 265. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 266. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 267. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 268. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 269. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 270. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 271. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 272. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 273. In another embodiment, the at least two immunogenic fragments each comprise, consist of, or consist essentially of an amino acid sequence represented by SEQ ID NO: 274.

(24) In certain embodiments, each of the at least two immunogenic fragments comprise, consist of, or consist essentially of a different amino acid sequence independently selected from the group consisting of SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), e.g., when the recombinant polypeptide comprises, consists of, or consists essentially of two immunogenic fragments, one immunogenic fragment comprises, consists of, or consists essentially of SEQ ID NO: 245 and the second immunogenic fragment comprises, consists of, or consists essentially of an amino acid sequence selected from the group consisting of SEQ ID NOs: 246-254.

(25) In some embodiments, the recombinant polypeptide includes a plurality of identical immunogenic fragments. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of the same amino acid sequence. In some embodiments, the recombinant polypeptide includes two, three, four or five identical immunogenic fragments. In yet further embodiments, the recombinant polypeptide includes two or three identical immunogenic fragments.

(26) In some embodiments, the recombinant polypeptide includes a plurality of non-identical immunogenic fragments. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of a different amino acid sequence from each other fragment, i.e., each fragment is a different fragment. In some embodiments, the recombinant polypeptide includes two, three, four or five different immunogenic fragments. In yet further embodiments, the recombinant polypeptide includes two or three different immunogenic fragments.

(27) In some embodiments, the recombinant polypeptide includes a plurality of immunogenic fragments, in which some of the fragments are identical, but not all. For example, a recombinant polypeptide can include two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of the same amino acid sequence, while also including one, two, three, four, five, six, seven, eight, nine, ten, or more immunogenic fragments, wherein each fragment comprises, consists of, or consists essentially of a different amino acid sequence from each other fragment. In some embodiments, the recombinant polypeptide includes a total of two, three, four or five immunogenic fragments. In yet further

embodiments, the recombinant polypeptide includes a total of two or three immunogenic fragments.

(28) In certain embodiments wherein the recombinant polypeptide comprises an antibody Fc region, the at least one immunogenic fragment can be arranged in any suitable serial orientation with respect to the Fc region. In some embodiments, the at least one immunogenic fragment is connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]_x-[N-terminus-Fc region-C-terminus], wherein X is an integer 1-10 representing the number of immunogenic fragments within the recombinant polypeptide. In some embodiments, the at least one immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]_x, wherein X is an integer 1-10 representing the number of immunogenic fragments within the recombinant polypeptide. In some embodiments, wherein the recombinant polypeptide includes at least two immunogenic fragments, at least one immunogenic fragment is connected to the N-terminus of the Fc region, and at least one immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]_x-[N-terminus-Fc region-C-terminus]-[immunogenic fragment]_y, wherein X and Y are independently an integer 1-10 representing the number of immunogenic fragments connected to each side of the Fc region. In some embodiments, wherein the recombinant polypeptide includes two immunogenic fragments, both immunogenic fragments are connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]-[immunogenic fragment]-[N-terminus-Fc region-C-terminus]. In other embodiments, wherein the recombinant polypeptide includes two immunogenic fragments, both immunogenic fragments are connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]-[immunogenic fragment]. In some embodiments, wherein the recombinant polypeptide includes three immunogenic fragments, each immunogenic fragment is connected to the N-terminus of the Fc region. This orientation can be depicted as [immunogenic fragment]-[immunogenic fragment]-[immunogenic fragment]-[N-terminus-Fc region-C-terminus]. In other embodiments, wherein the recombinant polypeptide includes three immunogenic fragments, each immunogenic fragment is connected to the C-terminus of the Fc region. This orientation can be depicted as [N-terminus-Fc region-C-terminus]-[immunogenic fragment]-[immunogenic fragment]-[immunogenic fragment].

(29) Embodiments of the recombinant polypeptide that include a plurality of immunogenic fragments provide for a flexible expression platform with robust expression of full-length protein, modality to modify individual or multiple domains within one or more of the plurality of immunogenic fragments to reflect the most recent virus variant sequence(s), allow for single step affinity purification by use of an Fc region if present in the recombinant polypeptide, and provide high-level, long-term immune response as tested in Rhesus macaques.

(30) In some embodiments, wherein the recombinant polypeptide includes a plurality of immunogenic fragments, the immunogenic fragments are connected to each other via a linker. The linker can be any suitable linker. Suitable linkers include a polypeptide comprising, consisting of, or consisting essentially of, an amino acid sequence of 1-35 residues, wherein each residue is independently serine, glycine, or aspartic acid, and further wherein the amino acid sequence contains zero or one aspartic acid residues. Other suitable linkers include a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long). When a recombinant polypeptide includes two or more such linkers, the amino acid sequence of the linkers can be identical or different. In some embodiments, wherein the recombinant polypeptide includes a plurality of immunogenic fragments, the immunogenic fragments are connected directly to each other without an intervening linker.

(31) In some embodiments, the recombinant polypeptide comprising or consisting essentially of

one or more immunogenic fragments further comprises an antibody Fc region. In some embodiments, the recombinant polypeptide comprising, consisting of, or consisting essentially of one or more immunogenic fragments does not comprise an antibody Fc region. In some embodiments, the recombinant polypeptide comprising or consisting essentially of the one or more immunogenic fragments initially comprises an antibody Fc region, e.g., the recombinant polypeptide comprises an antibody Fc region upon expression of the recombinant polypeptide, wherein the antibody Fc region is subsequently removed, for instance, enzymatically removed. Suitable enzymes include, but are not limited to, class I proteases such as IdeS and IdeZ, and class II enzymes such as the Tobacco Etch Virus (TEV), Rhinovirus 3C (Rv3C), and Factor Xa. When a class II enzyme is employed, the recombinant polypeptide comprises the cleavage site for the employed enzyme. In some embodiments, the cleavage site is within or adjacent to the linker region between the Spike protein subunit and the Fc domain.

(32) In some embodiments, removal occurs during a purification step or after the recombinant polypeptide has undergone at least one purification step. For instance, the purification is any suitable type of affinity purification, such as Protein A, Protein G, or Protein M purification. In some embodiments, subtractive chromatography steps are used to remove the enzyme. In subtractive chromatography, unwanted proteins are retained on the resin whilst desired proteins flow-through and are collected for further uses. Examples of suitable subtractive chromatography methods include IMAC for His6-tagged enzymes, and Protein A resin, Protein G resin, or Protein M resin to remove Fc-fusion enzymes. In some embodiments, cut Fcs and uncut Fc-fusions protein are removed by one or more of Protein A, Protein G, and Protein M resin chromatography.

(33) In some embodiments, the one or more immunogenic fragments are connected to the antibody Fc region via a linker. The linker can be any suitable linker. Suitable linkers include a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine, glycine, or aspartic acid, and further wherein the amino acid sequence contains zero or one aspartic acid residues. Other suitable linkers include a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long). When a recombinant polypeptide includes two or more such linkers, the amino acid sequence of the linkers can be identical or different from each other. In some embodiments, wherein the recombinant polypeptide includes at least one immunogenic fragment connected to the N-terminus of the Fc region, and/or at least one immunogenic fragment is connected to the C-terminus of the Fc region, the immunogenic fragment(s) nearest to the Fc region are connected directly to the Fc region without an intervening linker.

(34) The antibody Fc region included in the recombinant polypeptide can be any suitable antibody Fc region. Suitable antibody Fc regions include wild-type human or other animal Immunoglobulin Fc regions such as IgG, IgA, IgD, IgE, IgM, and their respective subclasses, for example, human IgG1 Fc regions and Fc regions derived therefrom. Other suitable antibody Fc regions include mutant Fc regions that enhance or diminish Fc-receptor binding affinity to speed up or slow down uptake, respectively. In some embodiments, the antibody Fc region comprises the amino acid sequence of SEQ ID NO: 71.

(35) In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 75 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 77 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 79 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 81 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 83 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 85 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-His8]), SEQ ID NO: 87 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 89 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 91 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID

NO: 93 (LS3487 [NTD_long_a-Rv3C-Fc]), SEQ ID NO: 95 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 97 (LS2354 [CTD_long_a D614G-Fc]), SEQ ID NO: 99 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 101 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 103 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 105 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 107 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 109 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 111 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 113 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 115 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 117 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 119 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 125 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 127 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 133 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 135 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 137 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 139 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 141 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 143 (LS2377 [RBD_a-Fc]), SEQ ID NO: 145 (LS2378 [RBD_b-Fc]), SEQ ID NO: 147 (LS2379 [RBD_c-Fc]), SEQ ID NO: 149 (LS2380 [RBD_d-Fc]), SEQ ID NO: 151 (LS2381 [RBD_e-Fc]), SEQ ID NO: 153 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 155 (LS2393 [(RBD-tight).sub.2-Fc]), SEQ ID NO: 157 (LS2394 [(RBD-extended).sub.2-Fc]), SEQ ID NO: 159 (LS2395 [(RBD).sub.2-Fc]), SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod. 1], SEQ ID NO: 167 (LS2423 [(CTD_short_i).sub.2-Fc)-mod. 2], SEQ ID NO: 169 (LS2435 [(CTD_short_i).sub.2-Fc)-mod. 3], SEQ ID NO: 217, SEQ ID NO: 219, SEQ ID NO: 221, SEQ ID NO: 223, SEQ ID NO: 225, SEQ ID NO: 227, SEQ ID NO: 229, SEQ ID NO: 231, SEQ ID NO: 233, SEQ ID NO: 235, SEQ ID NO: 237, SEQ ID NO: 237, SEQ ID NO: 239, SEQ ID NO: 241, SEQ ID NO: 243, SEQ ID NOs: 255-264 (BA.X-1-monomer-Fc-BA.X-10-monomer-Fc), and SEQ ID NOs: 275-284 (BA.X-1-dimer-Fc-BA.X-10-dimer-Fc). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod. 1], SEQ ID NO: 167 (LS2423 [(CTD_short_i).sub.2-Fc)-mod. 2], and SEQ ID NO: 169 (LS2435 [(CTD_short_i).sub.2-Fc)-mod. 3]. In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]). In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 163. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the amino acid sequence represented by SEQ ID NO: 169. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264 (BA.X-1-monomer-Fc-BA.X-10-monomer-Fc). In another embodiment, the recombinant polypeptide

[illegible]

91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 280. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 281. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 282. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 283. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence represented by SEQ ID NO: 284.

(36) In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NOs: 73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, 119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 255-264, and 275-284. In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 167 (LS2423 [(CTD_short_i).sub.2-Fc]), and SEQ ID NO: 169 (LS2435 [(CTD_short_i).sub.2-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence of SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]). In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]). In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence represented by SEQ ID NO: 163. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, the amino acid sequence represented by SEQ ID NO: 169. In some embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264 and SEQ ID NOs: 275-284. In further embodiments, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 255. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 256. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 257. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 258. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 259. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 260. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 261. In another embodiment, the recombinant polypeptide comprises, consists of,

or consists essentially of, an amino acid sequence represented by SEQ ID NO: 262. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 263. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 264. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 275. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 276. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 277. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 278. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 279. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 280. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 281. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 282. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 283. In another embodiment, the recombinant polypeptide comprises, consists of, or consists essentially of, an amino acid sequence represented by SEQ ID NO: 284.

(37) In some embodiments, the recombinant polypeptide is encoded by a recombinant polynucleotide. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 76 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 78 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 80 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 82 (LS3489 [CTD_short_a-Rv3c-Fc]), SEQ ID NO: 84 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 86 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-His8]), SEQ ID NO: 88 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 90 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 92 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 94 (LS3487 [NTD_long_a-Rv3c-Fc]), SEQ ID NO: 96 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 98 (LS2354 [CTD_long_a-D614G-Fc]), SEQ ID NO: 100 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 102 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 104 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 106 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 108 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 110 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 112 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 114 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 116 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 118 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 120 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 126 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 128 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 134 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 136 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 138 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 140 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 142 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 144 (LS2377 [RBD_a-Fc]), SEQ ID NO: 146 (LS2378 [RBD_b-Fc]), SEQ ID NO: 148 (LS2379 [RBD_c-Fc]), SEQ ID NO: 150 (LS2380 [RBD_d-Fc]), SEQ ID NO: 152 (LS2381 [RBD_e-Fc]), SEQ ID NO: 154 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 156 (LS2393 [(RBD-tight).sub.2-Fc]), SEQ ID NO: 158 (LS2394 [(RBD-extended).sub.2-Fc]), SEQ ID NO: 160 (LS2395 [(RBD).sub.2-Fc]), SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 168 (LS2423 [(CTD_short_i).sub.2-Fc]), SEQ ID

NO: 170 (LS2435 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 218, SEQ ID NO: 220, SEQ ID NO: 222, SEQ ID NO: 224, SEQ ID NO: 226, SEQ ID NO: 228, SEQ ID NO: 230, SEQ ID NO: 232, SEQ ID NO: 234, SEQ ID NO: 236, SEQ ID NO: 238, SEQ ID NO: 240, SEQ ID NO: 242, SEQ ID NO: 244, SEQ ID NOs: 295-304, and SEQ ID NOs: 315-324.

(38) In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod.1]), SEQ ID NO: 168 (LS2423 [(CTD_short_i).sub.2-Fc)-mod. 2]), and SEQ ID NO: 170 (LS2435 [(CTD_short_i).sub.2-Fc-mod. 3]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an nucleic acid sequence selected from the group consisting of SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]). In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence represented by SEQ ID NO: 164. In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence represented by SEQ ID NO: 170. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304 and SEQ ID NOs: 315-324. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to the nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304.

(39) In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 295-304, and 315-324. In some embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod. 1]), SEQ ID NO: 168 (LS2423 [(CTD_short_i).sub.2-Fc)-mod.2]), and SEQ ID NO: 170 (LS2435 [(CTD_short_i).sub.2-Fc)-mod. 3]). In some embodiments, the polynucleotide comprises, consists of, or consists essentially of, the nucleic acid sequence of SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]). In some embodiments, polynucleotide comprises, consists of, or consists essentially of, an nucleic acid sequence selected from the group consisting of SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), and SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]). In an embodiment, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence represented by SEQ ID NO: 164. In an embodiment, the polynucleotide comprises,

consists of, or consists essentially of a nucleic acid sequence represented by SEQ ID NO: 170. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304 and SEQ ID NOs: 315-324. In certain embodiments, the polynucleotide comprises, consists of, or consists essentially of a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 295-304.

(40) Nucleic acid or amino acid sequence “identity,” as referenced herein, can be determined by comparing a nucleic acid or amino acid sequence of interest to a reference nucleic acid or amino acid sequence. The percent identity is the number of nucleotides or amino acid residues that are the same (i.e., that are identical) as between the optimally aligned sequence of interest and the reference sequence divided by the length of the longest sequence (i.e., the length of either the sequence of interest or the reference sequence, whichever is longer). Alignment of sequences and calculation of percent identity can be performed using available software programs. Examples of such programs include CLUSTAL-W, T-Coffee, and ALIGN (for alignment of nucleic acid and amino acid sequences), BLAST programs (e.g., BLAST 2.1, BL2SEQ, BLASTp, BLASTn, and the like) and FASTA programs (e.g., FASTA3x, FASTM, and SSEARCH) (for sequence alignment and sequence similarity searches). Sequence alignment algorithms also are disclosed in, for example, Altschul et al., *J. Molecular Biol.*, 215(3): 403-410 (1990), Beigert et al., *Proc. Natl. Acad. Sci. USA*, 106(10): 3770-3775 (2009), Durbin et al., eds., *Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids*, Cambridge University Press, Cambridge, UK (2009), Soding, *Bioinformatics*, 21(7): 951-960 (2005), Altschul et al., *Nucleic Acids Res.*, 25(17): 3389-3402 (1997), and Gusfield, *Algorithms on Strings, Trees and Sequences*, Cambridge University Press, Cambridge UK (1997)). Percent (%) identity of sequences can be also calculated, for example, as $100 \times [(\text{identical positions}) / \min(\text{TG.sub.A}, \text{TG.sub.B})]$, where TG.sub.A and TG.sub.B are the sum of the number of residues and internal gap positions in peptide sequences A and B in the alignment that minimizes TG.sub.A and TG.sub.B. See, e.g., Russell et al., *J. Mol. Biol.*, 244: 332-350 (1994).

(41) In some embodiments, the nucleotide sequence of the polynucleotide is codon optimized and/or codon pair optimized.

(42) Another embodiment is a recombinant vector that comprises, consists of, or consists essentially of, a polynucleotide that encodes the recombinant polypeptide described herein. The recombinant vector can be any suitable vector. Examples of suitable recombinant vectors include but are not limited to a pcDNA3.1, a pSV, a pCMV, a pBApo-CMV, or a pBApo-EF1alpha expression vector.

(43) Yet another embodiment is an isolated cell that includes the recombinant polypeptide described herein or a recombinant polynucleotide that contains a nucleic acid sequence that encodes the recombinant polypeptide.

(44) Another embodiment is a pharmaceutical composition that contains the recombinant polypeptide described herein and at least one pharmaceutically acceptable carrier. The at least one pharmaceutically acceptable carrier can be any suitable carrier. Examples of suitable carriers include water and any suitable buffer. Suitable buffers include HEPES-buffered saline and phosphate-buffered saline. In certain embodiments, the pharmaceutical composition further contains at least one adjuvant. The at least one adjuvant can be any suitable adjuvant. Examples of suitable adjuvants include alum adjuvants, emulsion adjuvants, and pattern recognition receptor agonist adjuvants. Further examples include AS03, MF59, Squalene Emulsion, Alum, aluminum hydroxide gels, calcium phosphate hydroxide, paraffin oil, cytokines (IL-1, IL-2, IL-12), killed bacterial products such as *Bordetella* and *Mycobacterium* bacteria, bacterial toxoids, squalene and DL- α -tocopherol emulsions, squalene-oil-in-water emulsion, aluminum phosphate gels, saponins, cyclic dinucleotides, and TLR agonists, preferably TLR1, TLR2, TLR4, TLR5, TLR7, TLR8, TLR9 etc., and combinations thereof. In certain embodiments, the adjuvant is AS03. In certain other embodiments, the adjuvant is Alum. In yet further embodiments, the pharmaceutical

composition does not contain an adjuvant.

(45) The pharmaceutical composition can contain any therapeutically effective amount of the recombinant polypeptide described herein. A therapeutically effective amount is an amount sufficient to induce an immune response against the target virus or viruses, for instance, SARS-CoV-2, SARS-CoV, and/or MERS-CoV, preferably SARS-CoV-2. Typically, a dosage is therapeutically effective if it prevents, inhibits, reduces, eliminates, protects against, and/or delays the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject. Infectious clinical conditions include, for example, fever or chills, cough, shortness of breath or difficult breathing, fatigue, muscle or body aches, headache, loss of taste or smell, sore throat, congestion, runny nose, nausea, vomiting, and diarrhea. In some embodiments, a single dose of the pharmaceutical composition contains 10 nanograms to 1 milligram, 0.1-250 micrograms, 10-100 micrograms, or 12.5-50 micrograms of the recombinant polypeptide. In some embodiments, a single dose of the pharmaceutical composition contains 0.01-1, 0.1-1, 0.5-5, 1-20, or 5-15, 1-50, or 10-50 micrograms of the recombinant polypeptide. In some embodiments, wherein the pharmaceutical composition does not contain an adjuvant, a single dose of the pharmaceutical composition could contain 0.01-1, 0.1-1, 0.5-5, 1-20, or 5-15, 1-50, or 10-50 micrograms of the recombinant polypeptide. However, a single dose of the pharmaceutical composition could also contain increased amounts of the recombinant polypeptide, such as 100-1000, 100-250, or 250-500 micrograms of recombinant polypeptide.

(46) A further embodiment is a method for preventing, inhibiting, reducing, eliminating, protecting against, or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide described herein, the polypeptide encoded by the recombinant polynucleotide described herein, or a dose of the pharmaceutical composition described herein. Examples of beta coronaviruses include SARS-CoV, MERS-CoV, and SARS-CoV-2.

(47) Another embodiment is a method for inducing an immune response against a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide described herein, the polypeptide encoded by the recombinant polynucleotide described herein, or a dose of the pharmaceutical composition described herein. Examples of beta coronaviruses include SARS-CoV, MERS-CoV, and SARS-CoV-2.

(48) In the methods described herein, administration can be achieved by any suitable administration method. Suitable administration methods include oral, parenteral, subcutaneous, intravenous, intramuscular, intrapulmonary, intranasal, intraarterial, intrathecal, and intraperitoneal administration.

(49) In the methods described herein, the subject can be any suitable animal that is capable of being infected by a beta coronavirus, such as SARS-CoV, MERS-CoV, and SARS-CoV-2. The subject can be a human, non-human primate, horse, pig, cattle, cat, dog, sheep, mink, rodent, hamster, or bat, preferably human.

Examples of Non-Limiting Aspects of the Disclosure

(50) Aspects, including embodiments, of the invention described herein may be beneficial alone or in combination, with one or more other aspects or embodiments. Without limiting the foregoing description, certain non-limiting aspects of the disclosure numbered (1)-(36) are provided below. As will be apparent to those of skill in the art upon reading this disclosure, each of the individually numbered aspects may be used or combined with any of the preceding or following individually numbered aspects. This is intended to provide support for all such combinations of aspects and is not limited to combinations of aspects explicitly provided below:

(51) (1) A recombinant polypeptide comprising at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein and optionally an antibody Fc region.

(52) (2) The recombinant polypeptide of aspect 1, wherein the at least one fragment of the SARS-

CoV-2 spike glycoprotein comprises the N-terminal domain of the S1 subunit, the C-terminal domain of the S1 subunit, or both.

(53) (3) The recombinant polypeptide of aspect 1 or 2, wherein the at least one fragment of the SARS-CoV-2 spike glycoprotein comprises an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight).sub.2), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended).sub.2), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD).sub.2), SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3), SEQ ID NOS: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOS: 265-274 (BA.X-1-dimer-BA.X-10-dimer).

(54) (4) The recombinant polypeptide of any one of aspects 1-3, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274.

(55) (5) The recombinant polypeptide of any of one aspects 1-4, wherein the polypeptide comprises at least two immunogenic fragments.

(56) (6) The recombinant polypeptide of aspect 5, wherein each of the at least two immunogenic fragments comprises an amino acid sequence with at least 90%, preferably at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 1 (CTD_long_a), SEQ ID NO: 3 (CTD_long_a D614G), SEQ ID NO: 5 (CTD_long_a-Dimer), SEQ ID NO: 7 (CTD_long_b), SEQ ID NO: 9 (CTD_long_c), SEQ ID NO: 11 (CTD_long_d), SEQ ID NO: 13 (CTD_long_e), SEQ ID NO: 15 (CTD_long_f), SEQ ID NO: 17 (CTD_long_g), SEQ ID NO: 19 (CTD_long_h), SEQ ID NO: 21 (CTD_long_i), SEQ ID NO: 23 (CTD_short_a), SEQ ID NO: 25 (CTD_short_b), SEQ ID NO: 27 (CTD_short_c), SEQ ID NO: 29 (CTD_short_d), SEQ ID NO: 31 (CTD_short_e), SEQ ID NO: 33 (CTD_short_f), SEQ ID NO: 35 (CTD_short_g), SEQ ID NO: 37 (CTD_short_h), SEQ ID NO: 39 (CTD_short_i), SEQ ID NO: 41 (CTD_vs_a), SEQ ID NO: 43 (CTD_vs_b), SEQ ID NO: 45 (CTD_vs_c), SEQ ID NO: 47 (CTD_vs_d), SEQ ID NO: 49 (CTD_vs_e), SEQ ID NO: 51 (RBD_a), SEQ ID NO: 53 (RBD_b), SEQ ID NO: 55 (RBD_c), SEQ ID NO: 57 (RBD_d), SEQ ID NO: 59 (RBD_e), SEQ ID NO: 61 (NTD_long_a), SEQ ID NO: 63 (NTD_short_a), SEQ ID NO: 171 (RBD-tight), SEQ ID NO: 173 ((RBD-tight).sub.2), SEQ ID NO: 175 (RBD-extended), SEQ ID NO: 177 ((RBD-extended).sub.2), SEQ ID NO: 179 (RBD), SEQ ID NO: 181 ((RBD).sub.2), SEQ ID NO: 183 ((CTD_short_d).sub.2), SEQ ID NO: 185 ((CTD_short_i).sub.2), SEQ ID NO: 187 ((CTD_short_i).sub.2—mod. 1), SEQ ID NO: 189 ((CTD_short_i).sub.2—mod. 2), SEQ ID NO: 191 ((CTD_short_i).sub.2—mod. 3), SEQ ID NO: 199 (SARS-2003, SARS_short_h), SEQ ID NO: 201 (SARS-2003, SARS_short_i), SEQ ID NO: 203 (MERS_Lytic_a), SEQ ID NO: 205 (MERS_Lytic_b), SEQ ID NO: 207 (MERS_Lytic_c), SEQ ID NO: 209 (MERS_Lytic_d), SEQ ID

NO: 211 (MERS_Lytic_e), SEQ ID NO: 213 (MERS_Lytic_f), SEQ ID NO: 215 (MERS_Lytic_g), SEQ ID NOs: 245-254 (BA.X-1-BA.X-10), and SEQ ID NOs: 265-274 (BA.X-1-dimer-BA.X-10-dimer).

(57) (7) The recombinant polypeptide of aspect 5, wherein each of the at least two immunogenic fragments comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 171, 173, 175, 177, 179, 181, 183, 185, 187, 189, 191, 199, 201, 203, 205, 207, 209, 211, 213, 215, 245-254, and 265-274.

(58) (8) The recombinant polypeptide of any one of aspects 5-7, wherein each immunogenic fragment of the at least two immunogenic fragments comprises the same amino acid sequence.

(59) (9) The recombinant polypeptide of any one of aspects 5-8, wherein each immunogenic fragment of the at least two immunogenic fragments comprises a different amino acid sequence from the other immunogenic fragments.

(60) (10) The recombinant polypeptide of any one of aspects 1-9, wherein the polypeptide comprises two, three, four, or five immunogenic fragments.

(61) (11) The recombinant polypeptide of any one of aspects 5-10, wherein the at least two immunogenic fragments are connected to each other via a linker.

(62) (12) The recombinant polypeptide of aspect 11, wherein the linker is a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine or glycine.

(63) (13) The recombinant polypeptide of any one of aspects 1-12, wherein the at least one immunogenic fragment of the SARS-CoV-2 spike glycoprotein is connected to the antibody Fc region via a linker.

(64) (14) The recombinant polypeptide of aspect 13, wherein the linker comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 65 (Fc1), SEQ ID NO: 67 (Fc1-TEV), SEQ ID NO: 69 (Fc1-Rv3C), SEQ ID NO: 193 (Short), SEQ ID NO: 195 (Medium), and SEQ ID NO: 197 (Long).

(65) (15) The recombinant polypeptide of any one of aspects 1-14, wherein the antibody Fc region is from a human IgG1 antibody or derived therefrom.

(66) (16) The recombinant polypeptide of aspect 15, wherein the antibody Fc region comprises the amino acid sequence of SEQ ID NO: 71.

(67) (17) The recombinant polypeptide of any one of aspects 1-16, wherein the polypeptide comprises an amino acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 75 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 77 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 79 (LS3485 [CTD_long_a-TEV-Fc]), SEQ ID NO: 81 (LS3489 [CTD_short_a_Rv3c-Fc]), SEQ ID NO: 83 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 85 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-Hiss]), SEQ ID NO: 87 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 89 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 91 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 93 (LS3487 [NTD_long_a-Rv3C-Fc]), SEQ ID NO: 95 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 97 (LS2354 [CTD_long_a D614G-Fc]), SEQ ID NO: 99 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 101 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 103 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 105 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 107 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 109 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 111 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 113 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 115 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 117 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 119 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 121 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 123 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 125 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 127 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 129 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 131 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 133 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 135 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 137 (LS2374

[CTD_vs_c-Fc]], SEQ ID NO: 139 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 141 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 143 (LS2377 [RBD_a-Fc]), SEQ ID NO: 145 (LS2378 [RBD_b-Fc]), SEQ ID NO: 147 (LS2379 [RBD_c-Fc]), SEQ ID NO: 149 (LS2380 [RBD_d-Fc]), SEQ ID NO: 151 (LS2381 [RBD_e-Fc]), SEQ ID NO: 153 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 155 (LS2393 [(RBD-tight).sub.2-Fc]), SEQ ID NO: 157 (LS2394 [(RBD-extended).sub.2-Fc]), SEQ ID NO: 159 (LS2395 [(RBD).sub.2-Fc]), SEQ ID NO: 161 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 163 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 165 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod. 1], SEQ ID NO: 167 (LS2423 [(CTD_short_i).sub.2-Fc)-mod. 2], SEQ ID NO: 169 (LS2435 [(CTD_short_i).sub.2-Fc)-mod. 3], SEQ ID NOs: 255-264, and SEQ ID NOs: 275-284.

(68) (18) The recombinant polypeptide of aspect 17, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NOS: 73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, 119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 255-264, and 275-284.

(69) (19) A recombinant polynucleotide encoding the recombinant polypeptide of any one of aspects 1-18.

(70) (20) The recombinant polynucleotide of aspect 19, wherein the polynucleotide comprises a nucleic acid sequence with at least 90%, preferably at least 95%, 96%, 97%, 98%, or 99%, sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74 (LS2330 [CTD_short_a-Fc]), SEQ ID NO: 76 (LS3472, LS3473, LS3474 [CTD_long_a-Fc]), SEQ ID NO: 78 (LS3477 [CTD_short_a-TEV-Fc]), SEQ ID NO: 80 (LS3485 [CTD_long_a TEV-Fc]), SEQ ID NO: 82 (LS3489 [CTD_short_a_Rv3c-Fc]), SEQ ID NO: 84 (LS3497 [CTD_long_a-Rv3c-Fc]), SEQ ID NO: 86 (LS2316, LS2317, LS2318, LS2319 [CTD_long_a-Hiss]), SEQ ID NO: 88 (LS3479 [NTD_short_a-TEV-Fc]), SEQ ID NO: 90 (LS3475 [NTD_long_a-TEV-Fc]), SEQ ID NO: 92 (LS3491 [NTD_short_a-Rv3c-Fc]), SEQ ID NO: 94 (LS3487 [NTD_long_a-Rv3C-Fc]), SEQ ID NO: 96 (LS2326 [NTD_long_a-Fc]), SEQ ID NO: 98 (LS2354 [CTD_long_a D614G-Fc]), SEQ ID NO: 100 (LS2355 [CTD_long_a-Dimer-Fc]), SEQ ID NO: 102 (LS2356 [CTD_long_b-Fc]), SEQ ID NO: 104 (LS2357 [CTD_long_c-Fc]), SEQ ID NO: 106 (LS2358 [CTD_long_d-Fc]), SEQ ID NO: 108 (LS2359 [CTD_long_e-Fc]), SEQ ID NO: 110 (LS2360 [CTD_long_f-Fc]), SEQ ID NO: 112 (LS2361 [CTD_long_g-Fc]), SEQ ID NO: 114 (LS2362 [CTD_long_h-Fc]), SEQ ID NO: 116 (LS2363 [CTD_long_i-Fc]), SEQ ID NO: 118 (LS2364 [CTD_short_b-Fc]), SEQ ID NO: 120 (LS2365 [CTD_short_c-Fc]), SEQ ID NO: 122 (LS2366 [CTD_short_d-Fc]), SEQ ID NO: 124 (LS2367 [CTD_short_e-Fc]), SEQ ID NO: 126 (LS2368 [CTD_short_f-Fc]), SEQ ID NO: 128 (LS2369 [CTD_short_g-Fc]), SEQ ID NO: 130 (LS2370 [CTD_short_h-Fc]), SEQ ID NO: 132 (LS2371 [CTD_short_i-Fc]), SEQ ID NO: 134 (LS2372 [CTD_vs_a-Fc]), SEQ ID NO: 136 (LS2373 [CTD_vs_b-Fc]), SEQ ID NO: 138 (LS2374 [CTD_vs_c-Fc]), SEQ ID NO: 140 (LS2375 [CTD_vs_d-Fc]), SEQ ID NO: 142 (LS2376 [CTD_vs_e-Fc]), SEQ ID NO: 144 (LS2377 [RBD_a-Fc]), SEQ ID NO: 146 (LS2378 [RBD_b-Fc]), SEQ ID NO: 148 (LS2379 [RBD_c-Fc]), SEQ ID NO: 150 (LS2380 [RBD_d-Fc]), SEQ ID NO: 152 (LS2381 [RBD_e-Fc]), SEQ ID NO: 154 (LS2382 [NTD_short_a-Fc]), SEQ ID NO: 156 (LS2393 [(RBD-tight).sub.2-Fc]), SEQ ID NO: 158 (LS2394 [(RBD-extended).sub.2-Fc]), SEQ ID NO: 160 (LS2395 [(RBD).sub.2-Fc]), SEQ ID NO: 162 (LS2397-2400 [(CTD_short_d).sub.2-Fc]), SEQ ID NO: 164 (LS2401-2404 [(CTD_short_i).sub.2-Fc]), SEQ ID NO: 166 (LS2421 and LS2422 [(CTD_short_i).sub.2-Fc)-mod. 1], SEQ ID NO: 168 (LS2423 [(CTD_short_i).sub.2-Fc)-mod. 1], and SEQ ID NO: 170 (LS2435 [(CTD_short_i).sub.2-Fc)-mod. 3], SEQ ID NOs: 295-304, and SEQ ID NOs: 315-324.

(71) (21) The recombinant polynucleotide of aspect 20, wherein the polynucleotide comprises a nucleic acid sequence selected from the group consisting of SEQ ID NO: 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128,

- 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 295-304, and 315-324.
- (72) (22) The recombinant polynucleotide of any one of aspects 19-21, wherein the nucleic acid sequence has been codon optimized.
- (73) (23) A pharmaceutical composition comprising the recombinant polypeptide of any one of aspects 1-18 or the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, and a pharmaceutically acceptable carrier.
- (74) (24) The pharmaceutical composition of aspect 23, wherein the pharmaceutical composition further comprises at least one adjuvant.
- (75) (25) The pharmaceutical composition of aspect 24, wherein the at least one adjuvant is selected from the group consisting of alum adjuvants, emulsion adjuvants, and pattern recognition receptor agonist adjuvants.
- (76) (26) The pharmaceutical composition of aspect 25, wherein the at least one adjuvant is MF59, Squalene Emulsion, Alum, aluminum hydroxide gels, calcium phosphate hydroxide, paraffin oil, cytokines (IL-1, IL-2, IL-12), killed bacterial products such as *Bordetella* and *Mycobacterium* bacteria, bacterial toxoids, squalene and DL- α -tocopherol emulsions, aluminum phosphate gels, saponins, cyclic dinucleotides, and TLR agonists, preferably TLR1, TLR2, TLR4, TLR5, TLR7, TLR8, TLR9 etc., and combinations thereof.
- (77) (27) The pharmaceutical composition of aspect 24, wherein the at least one adjuvant is a squalene-oil-in-water emulsion adjuvant.
- (78) (28) A vector comprising the recombinant polynucleotide of any one of aspects 19-22.
- (79) (29) An isolated cell comprising the recombinant polypeptide of any one of aspects 1-18, or the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22.
- (80) (30) A method for preventing, inhibiting, reducing, eliminating, protecting, or delaying the onset of an infection or an infectious clinical condition caused by a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27.
- (81) (31) A method for inducing an immune response against a beta coronavirus in a subject comprising administering to the subject the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27.
- (82) (32) The method of aspect 30 or 31, wherein the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 is administered by oral, parenteral, subcutaneous, intravenous, intramuscular, intranasal, intrapulmonary, intraarterial, intrathecal, or interperitoneal administration.
- (83) (33) The method of any one of aspects 30-32, wherein the coronavirus is selected from the group consisting of SARS-CoV-2, SARS-CoV, and MERS-CoV.
- (84) (34) The method of any one of aspects 30-33, wherein the subject is a mammal, preferably a human or non-human primate.
- (85) (35) The use of the recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 for the preparation of a medicament for the treatment or prevention of illness caused by SARS-CoV-2.
- (86) (36) The recombinant polypeptide of any one of aspects 1-18, the polypeptide encoded by the recombinant polynucleotide of any one of aspects 19-22, or the pharmaceutical composition of any one of aspects 23-27 for use as a medicament.

EXAMPLES

- (87) The following descriptions of cloning and protein expression apply to each example.

(88) Cloning: Inserts encoding a certain fragment of SARS CoV-2 virus Spike protein 51 were designed based prediction and previous Lytic Solutions expression data. Codon optimized cDNA was synthesized. The insert was cloned using either (a) restriction digests and DNA ligations or (b) NEB HiFi DNA assembly builder mix into a pcDNA3.1 vector containing appropriate secretion signal sequences, linkers, and tags for secreted fusion protein expression, as well as sequence encoding a human IgG1 Fc region. The resulting clone encoding a recombinant SARS CoV-2 Spike protein fragment-Fc region fusion protein (“recombinant CoV-2 fusion protein”) was verified by either colony PCR and/or restriction digests. DNA sequencing was also used.

(89) Protein Expression: Supercoiled plasmid of the verified clone was transiently transfected into CHO-S cells and expressed under the control of a constitutive promoter within cell culture conditions ranging from 30-37° C. and 3-10 days in CO.sub.2 (8%) incubators with rotary shaking agitation at speeds of 150 RPM. Cells were removed by centrifugation and culture medium containing the recombinant CoV-2 fusion protein were passed over Protein A agarose to bind the Fc region-containing CoV-2 protein. Filtration could also have been used to remove cells. The column containing bound recombinant CoV-2 fusion protein was washed with phosphate buffered saline. The recombinant CoV-2 fusion protein bound to the Protein A column was eluted with low pH glycine followed by neutralization in pH8.0 Tris. The recombinant CoV-2 fusion protein was dialyzed to remove glycine/tris and placed into HEPES-buffered saline (10 mM HEPES. 150 mM NaCl, pH adjusted with NaOH to pH 7.2-7.5). No additional purification was employed in this case. However, additional purification by any chromatography method such as HIC or ion exchange can optionally be used.

Example 1

(90) SARS CoV-2 antigens were selected from the genomic sequence (ncbi.nlm.nih.gov/nuccore/MN908947) to use in generating antibody and T-cell responses to the receptor binding domains of SARS CoV-2 virus Spike protein. The selected domain encoding regions were codon optimized for CHO cell expression using IDTDNA codon optimization algorithms. Template DNA was synthesized at Twist Bio. A modified pcDNA3.1 vector was used for protein expression that included human IgG1 Fc for translational fusion generation, i.e., a fusion protein containing a SARS CoV-2 virus Spike protein fragment and an antibody Fc region. The vector encoding the fusion protein was expressed in CHO cells, in which the fusion protein was secreted to the media, cells and cell debris were removed by centrifugation, the recombinant protein was captured with Protein A resin, and eluted with low pH glycine buffer. The resulting fusion protein was buffer exchanged by dialysis and mixed with adjuvant. The resulting vaccine composition was injected intramuscularly into Cynomolgus monkeys. The vaccine composition generated an unexpectedly strong immune response in Cynomolgus monkeys.

(91) Vaccination: Following protein expression in accordance with the description above, the purified recombinant CoV-2 fusion protein (LS2330, CTD_short-a-Fc; SEQ ID NO: 73) was mixed with Titermax Gold adjuvant, a modified squalene in water emulsion adjuvant, according to manufacturer directions, and injected intramuscularly into the thigh of Cynomolgus monkeys. Injections were performed at day 0 and day 14. Dosages were 250 ug of antigen of CTD_short-a-Fc (Seq ID NO:73). Prior to the first injection on day 0, a baseline sample was collected from each test subject.

(92) The immune response was monitored every two weeks following the injection on day 0, with the earliest sample taken on day 14. Accordingly, samples were collected on day 14 and day 28. The samples were analyzed, and the results are shown in FIG. 1. Venous blood was obtained from Cynomolgus monkeys pre immunization in EDTA-containing vacutubes (value shown as Day 0 serum sample in FIG. 1). Immediately after the pre-immunization blood sample was taken, monkeys were immunized with CTD_short-a-Fc (Seq ID NO:73). Serum was isolated by centrifugation of the non-coagulated blood. Serum was diluted 1:100 and analyzed on Intuitive Biosciences ELISA platform for anti-S1 spike binding antibodies. Additional time points were

taken at 2 weeks post Day 0, 4 weeks post Day 0, and 6 weeks post Day 0. The single immune boost (CTD_short-a-Fc (Seq ID NO:73)) was performed immediately after the 2 week blood sample was taken. Samples were processed similar to Day 0 samples and immune response was recorded as Relative Intensity Units (RIU) on Intuitive Bioscience ELISA platform. Capture of anti-SARS CoV-2 antibodies was performed by spotting SARS CoV-2 S1 spike protein (Sino Biologicals) onto the wells of Intuitive Bioscience ELISA platform 96 well plates, adding diluted serum—full concentration serum and dilutions to 1:200 serum:buffer were used, but higher concentrations of serum surprisingly resulted in a signal too strong to read on the platform, thus requiring a 1:200 fold dilution for the readings shown in FIG. 1—and incubating for 10-120 minutes to allow binding of anti-CTD antibodies to the S1 spike protein attached to the plate wells. Serum was washed 3× from the wells, and followed by application of anti-cynomolgus detection antibody to the well, incubated for 10-120 minutes, then washed 3× from the well. Detection reagents were added and the signal was quantified compared to control spots of unrelated proteins. (93) FIG. 1 depicts an unexpectedly strong immune response in Cynomolgus monkeys to the tested construct, particularly in comparison to the immune response generated by other known SARS-CoV-2 constructs. See, for instance, Graham et al., “Evaluation of the immunogenicity of prime-boost vaccination with the replication-deficient viral vectored COVID-19 vaccine candidate ChAdOx1 nCoV-19,” bioRxiv preprint doi: doi.org/10.1101/2020.06.20.159715 (posted Jun. 20, 2020)

(94) Neutralization Assays: FIGS. 5A-C depict the results of neutralization assays done on the tested Cynomolgus monkeys P0101 and P0102 (respectively depicted as “Animal 101” and “Animal 102” in FIG. 1). Regarding FIGS. 5A and 5B, “wpi” in the legend denotes weeks post injection/immunization. SARS-CoV-2 pseudotyped particles were generated as previously described (see Schmidt, F., et al. Measuring SARS-CoV-2 neutralizing antibody activity using pseudotyped and chimeric viruses. *J Exp Med*, v. 217, n. 11, 11 2020). Briefly, 293T cells were transfected with pHIV-1NLGagPol, pCCNG/nLuc and pSARS-CoV-2-SΔ19. Particles were harvested 48 hours after transfection, filtered and stored at −80° C. Fourfold serially diluted serum from the immunized monkeys was incubated with SARS-CoV-2 pseudotyped virus for 1 h at 37° C. The mixture was subsequently incubated with 293T/ACE2c1.22 cells (plated on Poly-D-Lysine-coated 96-well plates) with the final starting dilution of serum being 1:50. At 48 h later the cells were washed with PBS and lysed with Luciferase Cell Culture Lysis 5× reagent (Promega). Nanoluc Luciferase activity in lysates was measured using the Nano-Glo Luciferase Assay System (Promega) with the Modulus II Microplate Reader (Turner BioSystems). The raw nanoluc luciferase activity values (relative luminescence units) were normalized to those derived from cells infected with SARS-CoV-2 pseudotyped virus in the absence of serum or a rabbit monoclonal antibody diluted in normal human serum at 0.105 mg/mL (40592-R001, Sinobiological, Wayne, PA). The half-maximal inhibitory concentration for serum (NT50) was determined using four-parameter nonlinear regression (GraphPad Prism).

(95) Immunization of the two Cynomolgus macaques (IDs: P0101 and P0102) with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc] produced robust neutralizing antibody response. Neutralization assays were performed using a replication-defective single-cycle pseudotyped virus carrying SARS-CoV-2 spikes and the NanoLuc luciferase reporter. This assay has been previously shown to accurately predict serum neutralizing activity against authentic SARS-CoV-2 (see Schmidt, F., et al. referenced above). As a control for neutralization sensitivity, human serum obtained from a SARS-CoV-2 negative individual was used alone or spiked with a monoclonal neutralizing antibody (FIG. 5C). Serum samples collected at the various timepoints from 0 to 20 weeks post-immunization were evaluated for neutralizing activity. Sera from animals immunized with SEQ ID NO: 73 (LS2330 [CTD_short_a-Fc] had readily detectable neutralization activity, as early as 2 weeks post-immunization that significantly increased until weeks 4 to 8 of the study. Indeed, neutralizing titers were exceptionally high at 4-8 weeks after immunization, in the range of 10,000 to 100,000 and

were maintained in the 1000 to 10,000 range at 20 weeks after immunization.

Example 2

(96) Overview: SARS CoV-2 antigens were selected from the SARS CoV-2 genomic sequence (ncbi.nlm.nih.gov/nuccore/MN908947) to use in generating antibody and T-cell responses to the receptor binding domains of SARS CoV-2 virus Spike protein. Preliminary data identified select individual regions of the SARS CoV-2 virus Spike protein that would be amenable to high-level expression as Fc-fusions, that are resistant to proteolysis when expressed in CHO cells, and that generate strong immune response in Cynomolgus macaques. We further determined that in-series multimerization of select SARS CoV-2 virus Spike protein domains can be used to create proteins that retain high-level expression without significant proteolysis sensitivity whilst doubling the theoretical antigenicity of the molecule to be used for immune stimulation. The design of the multiple domain molecules provides a scaffold for straight-forward modification to incorporate amino acid mutations identified in new and emerging variants of the SARS CoV-2 virus Spike protein. The resulting vaccine composition of the Wuhan variant was injected intramuscularly into Rhesus macaques and elicit stronger immune responses using simple adjuvants at doses at mere fractions of that needed with single domain Fc-fusions.

(97) Following protein expression in accordance with the description above, resulting protein was analyzed by reducing SDS-PAGE to determine protein integrity for monomeric CTD-Fcs (FIGS. 2A-D), in-series dimer CTD-Fcs (FIG. 3) and in-series dimer mutant CTD-Fcs (FIG. 4). Protein yields were calculated using absorbance of 280 nm light and protein-specific extinction coefficients determined in silico according to their expected amino acid composition. Representative protein yields are given in Table 1.

(98) TABLE-US-00001 TABLE 1 Expression (mg protein/L CHO Strain(s) Insert culture ($\pm$ SD))

3472 (SEQ ID NO: 75) CTD_long_a (SEQ ID NO: 1)	77.3 .sup.(5)	2356 (SEQ ID NO: 101)
CTD_long_b (SEQ ID NO: 7)	60.8 .sup.(5)	2357 (SEQ ID NO: 103) CTD_long_c (SEQ ID NO: 9)
60.8 .sup.(5)	2358 (SEQ ID NO: 105) CTD_long_d (SEQ ID NO: 11)	65.8 .sup.(5)
2359 (SEQ ID NO: 107) CTD_long_e (SEQ ID NO: 13)	96.1 .sup.(5)	2360 (SEQ ID NO: 109) CTD_long_f (SEQ ID NO: 15)
95.1 .sup.(5)	2361 (SEQ ID NO: 111) CTD_long_g (SEQ ID NO: 17)	74.5 .sup.(5)
2362 (SEQ ID NO: 113) CTD_long_h (SEQ ID NO: 19)	108.3 .sup.(5)	2363 (SEQ ID NO: 115) CTD_long_i (SEQ ID NO: 21)
108.5 .sup.(5)	2330 (SEQ ID NO: 73) CTD_short_a (SEQ ID NO: 23)	>40 .sup.(5)
2364 (SEQ ID NO: 117) CTD_short_b (SEQ ID NO: 25)	131.8 .sup.(5)	2365 (SEQ ID NO: 119) CTD_short_c (SEQ ID NO: 27)
126.5 .sup.(5)	2366 (SEQ ID NO: 121) CTD_short_d (SEQ ID NO: 29)	106.2 .sup.(5)
2367 (SEQ ID NO: 123) CTD_short_e (SEQ ID NO: 31)	107.5 .sup.(5)	2368 (SEQ ID NO: 125) CTD_short_f (SEQ ID NO: 33)
109.9 .sup.(5)	2369 (SEQ ID NO: 127) CTD_short_g (SEQ ID NO: 35)	132.4 .sup.(5)
2370 (SEQ ID NO: 129) CTD_short_h (SEQ ID NO: 37)	67.5 .sup.(5)	2371 (SEQ ID NO: 131) CTD_short_i (SEQ ID NO: 39)
167.7 .sup.(5)	2372 (SEQ ID NO: 133) CTD_vs_a (SEQ ID NO: 41)	2.3 .sup.(5)
2373 (SEQ ID NO: 135) CTD_vs_b (SEQ ID NO: 43)	1.7 .sup.(5)	2374 (SEQ ID NO: 137) CTD_vs_c (SEQ ID NO: 45)
0.9 .sup.(5)	2375 (SEQ ID NO: 139) CTD_vs_d (SEQ ID NO: 47)	≤ 0.8 .sup.(5)
2376 (SEQ ID NO: 141) CTD_vs_e (SEQ ID NO: 49)	≤ 0.8 .sup.(5)	2377 (SEQ ID NO: 143) RBD_a (SEQ ID NO: 51)
≤ 0.8 .sup.(5)	2378 (SEQ ID NO: 145) RBD_b (SEQ ID NO: 53)	≤ 0.8 .sup.(5)
2380 (SEQ ID NO: 149) RBD_d (SEQ ID NO: 55)	≤ 0.8 .sup.(5)	2381 (SEQ ID NO: 151) RBD_e (SEQ ID NO: 57)
14.5 .sup.(5)	2393 (SEQ ID NO: 155) (RBD-tight).sub.2 (SEQ ID NO: 173)	≤ 0.8 .sup.(7)
2394 (SEQ ID NO: 157) (RBD-extended).sub.2 (SEQ ID NO: 177)	7.4 .sup.(7)	2395 ((SEQ ID NO: 159) (RBD).sub.2 (SEQ ID NO: 181)
≤ 0.8 .sup.(7)	2397-2400 (SEQ ID NO: 161) (CTD_short_d).sub.2 (SEQ ID NO: 183)	32.2 .sup.(4)/53.0 (± 4) .sup.(7)
2401-2404 (SEQ ID NO: 163) (CTD_short_i).sub.2 (SEQ ID NO: 185)	88.3 .sup.(4)/144.7 (± 5.5) .sup.(7)	2421 (SEQ ID NO: 165) [(CTD_short_i).sub.2-Fc; D2 mutations for 79.5 .sup.(4)
501.V2 variant, K417N, E484K, N501Y] (SEQ ID NO: 187)	2423 (SEQ ID NO: 167) [(CTD_short_i).sub.2-Fc; D2 mutations for 55.2 .sup.(4)	hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y]

(SEQ ID NO: 189) 2435 (SEQ ID NO: 169) [(CTD_short_i).sub.2-Fc; D1 and D2 mutations 61.2 .sup.(4) for hybrid P.1 and CAL.20C variants; K417T, L452R, E484K, N501Y] (SEQ ID NO: 191) .sup.(4) 4-day expression, .sup.(5) 5-day expression, .sup.(7) 7-day expression

(99) Based on the desire to develop methods to increase the relative antigen content of a protein molecule, in-series concatemers of RBD-containing protein fragments translationally fused to an Fc. CTD_short_d and CTD_short_i (SEQ ID NOs: 29 and 39, respectively) were prepared and selected for in-series expression given their high-level, protease resistant expression as monomers. When conjoined with a serine-glycine linker and expressed in CHO cells, the resulting double-domain proteins expressed at similar levels as the monomer constructs. In addition, the in-series design retained the molecular resistance to proteolysis during expression and purification. Therefore, by combining intramolecular dimerization driven by Fc interactions with in-series domain expression, it was possible to go from a single SARS-CoV-2 antigen fragment per molecule to having 4 (and potentially more) per molecule. The higher antigen content in conjunction with robust expression levels and protein stability provides a suitable framework for an immunogen to be used for vaccine and boost applications.

(100) As an example of this approach, double-domain Fc-fusion constructs (i.e., recombinant polypeptides containing two immunogenic fragments connected to an antibody Fc region) containing amino acid mutations within one or both of the domains were prepared. The mutations used were either from a single virus variant or a hybrid of two virus variants. Incorporation of the mutations into the double-domain-Fc wild-type molecule (strain 2401) depicted in FIG. 3 followed by expression in CHO cells resulted in intact, soluble protein with similar yields to the wild-type amino acid sequence. These results support the concept that this multi-domain Fc-fusion platform provides a robust scaffold for the incorporation and expression molecules that reflect new variant mutations. This provides a robust, timely and cost-effective system to adapt vaccine composition to meet the needs to mitigate evolving variants. A vaccine containing strain 2401 was subsequently tested in Rhesus macaque monkeys.

(101) Vaccination: Vaccination and boosters to nCoV-2 double dimer (Lytic Solutions Strain #2401, SEQ ID NO: 163, administered to all animals except for those indicated to be administered the nCoV-2 quadruple mutant vaccine) or nCoV-2 quadruple mutant vaccine (Lytic Solutions Strain #2435, SEQ ID NO: 169, referred to in Table 2C as “Variant COVID Vaccine” in Adjuvant & Dose column) were performed on Rhesus macaque monkeys by the following methodology. 50, 25, or 12.5 micrograms (these numbers are used to define dosage) of nCoV-2 protein was mixed with either AS03 (Invivogen catalog vac-as03-10) or alum (Invivogen catalog vac-alu-250) as a 1:1 volume mixture protein:adjuvant. For 50 ug doses 500 ul of each protein and adjuvant were used, for 25 ug and 12.5 ug 250 ul of each protein and adjuvant were used. Dosages were split and injected intramuscularly into left and right thighs at the time of initial vaccination and at booster vaccination. Initial vaccination day was designated day 0. Boosters were given 28 day post day 0 unless otherwise noted. Serum samples were taken prior to initial vaccination on day 0, day 14, day 28 prior to booster vaccination, and day 42 and day 56 were 2/4 weeks post booster injection. Therefore data from 14 and 28 (immunization plus 2 weeks and immunization plus 4 weeks, respectively) days are specific to a single dose of vaccine whereas samples from day 42 and day 56 are 2 doses of vaccine (booster plus 2 weeks, and booster plus 4 weeks respectively). Each animal received the same dosage of protein and adjuvant for the booster dose that they received in the primary dose.

(102) Analysis of antibody titers were performed at Intuitive Biosciences, Madison WI on their proprietary ELISA system (918 Deming Way, Suite 100, Madison WI 53719 USA). Serum was serially diluted in CSA buffer (Intuitive product no. 7-1037) to dilutions of 1:100, 1:1000, 1:10,000, 1:100,000, and 1:1,000,000. Analysis was performed at and by Intuitive Biosciences. ELISA units are measured as density on their platform with a maximum signal of approximately 45,000-50,000 counts. Titer signals were determined from dilutions that yielded signals less than

1/3rd maximal signal. Fifteen animals were used to determine vaccine/booster efficacy. Animal names are codes generated for each animal at the primate facility at UW-Madison. All animals were assayed through day 56 post initial vaccination. 50, 25, and 12.5 ug doses gave similar titers for day 28. Some animals were followed past the study design point of 56 days to various time points up to 23 weeks post initial vaccine (due to continued potency of vaccine response). Data is summarized in Tables 2A, 2B, and 2C. It is noted that, although some of the Plate Sample ID numbers are overlapping, e.g., Tables 2A and 2B both have a row with a plate sample ID of 59, this is merely an artifact of the data collection process, such that rows with the same Plate Sample ID represent independently collected data points.

(103) 50 ug, 25 ug, 12.5 ug: These data show that relatively low dosages of the tested constructs, including 12.5 ug, still elicited sufficient immune stimulation and/or boosting. Such low amounts of protein per vaccine dose allow for an increased number of active doses to be produced per liter of cell culture. As production levels increase, not only do cost of goods go down, but the timeframe to produce large numbers of doses can be decreased in comparison to vaccine compositions requiring higher amounts of protein per dose to provide sufficient immune protection. This is particularly relevant to providing immune protection against any newly-arising SARS-CoV-2 strains.

(104) The assembly of four point-mutations into a single RBD polypeptide was undertaken prior to public disclosure of the Delta mutant isolated first in India. Vaccine LS2435 (SEQ ID NO: 169) contains mutations in four sites that reflect mutation that evolved from new variants identified from the UK, South Africa, Brazil and southern California. Convergent evolution of mutations in new variants lead this to be an attractive approach of stacking multiple mutations in one construct to represent multiple variants, such as seen in the Delta variant. Each RBD point mutation was determined to add virulency through either immune system avoidance or enhanced viral entry or production of higher viral loads (or a combination thereof). The identified mutations were combined to develop a vaccine that could address each mutation and immune epitope singly or in combination. When evolutionary boundaries are considered, it became clear that mutational stacking was likely to take place through either recombination of previous viral strains, or additional mutations stacked onto previous viral strains that enhanced virulence/transmission. Considering the mutation-stacked SARS CoV-2 strains can evade the immune response as well as generate a more potent viral titer, having a vaccine that displays high levels of antibody and T-cell potency is a major advantage over previous COVID-19 vaccines which can provide less potent immune responses to these mutants. The SARS CoV-2 mutant-containing vaccine LS2435 demonstrates highly potent immune stimulation and antibody production in Rhesus macaques. In fact, the levels of antibodies achieved with the mutant vaccine are similar to responses seen from the tested wild-type vaccine, i.e., LS2401.

(105) TABLE-US-00002 TABLE 2A Animal - Plate Adjuvant & Days post Sample ID Spike S1
Spike S2 Nucleocapsid Dilution Dosage injection 59 14 217 296 1:100 BH56 - 0 AS03 50 ug 60
19 21 418 1:1,000 BH56 - 0 AS03 50 ug 61 0 30 100 1:10,000 BH56 - 0 AS03 50 ug 62 44982
343 489 1:100 BH56 - 14 AS03 50 ug 63 31862 49 445 1:1,000 BH56 - 14 AS03 50 ug 64
3227 28 276 1:10,000 BH56 - 14 AS03 50 ug 65 47053 507 642 1:100 BH56 - 28 AS03 50 ug
66 35647 42 46 1:1,000 BH56 - 28 AS03 50 ug 67 9818 57 290 1:10,000 BH56 - 28 AS03 50 ug
68 34 116 508 1:100 BC43 - 0 Alum 50 ug 69 0 30 291 1:1,000 BC43 - 0 Alum 50 ug 70 22 52
1 1:10,000 BC43 - 0 Alum 50 ug 71 36334 67 208 1:100 BC43 - 14 Alum 50 ug 72 14375 720
553 1:1,000 BC43 - 14 Alum 50 ug 73 234 0 0 1:10,000 BC43 - 14 Alum 50 ug 74 46731 20 36
1:100 BC43 - 28 Alum 50 ug 75 39535 56 258 1:1,000 BC43 - 28 Alum 50 ug 76 5458 17 1
1:10,000 BC43 - 28 Alum 50 ug 77 0 97 318 1:100 BH95 - 0 Alum 50 ug 78 0 66 146 1:1,000
BH95 - 0 Alum 50 ug 79 1 12 23 1:10,000 BH95 - 0 Alum 50 ug 80 42119 63 259 1:100 BH95 -
14 Alum 50 ug 81 30356 59 644 1:1,000 BH95 - 14 Alum 50 ug 82 451 0 0 1:10,000 BH95 - 14
Alum 50 ug 83 49046 156 63 1:100 BH95 - 28 Alum 50 ug 84 40830 0 187 1:1,000 BH95 - 28

Alum 50 ug 85 2782 28 78 1:10,000 BH95 - 28 Alum 50 ug 86 40 55 557 1:100 BG86 - 0 AS03 50 ug 87 40 93 766 1:1,000 BG86 - 0 AS03 50 ug 88 0 27 229 1:10,000 BG86 - 0 AS03 50 ug 89 27694 144 618 1:100 BG86 - 14 AS03 50 ug 90 436 114 275 1:1,000 BG86 - 14 AS03 50 ug 91 243 23 91 1:10,000 BG86 - 14 AS03 50 ug 92 46656 85 618 1:100 BG86 - 28 AS03 50 ug 93 38813 84 527 1:1,000 BG86 - 28 AS03 50 ug 94 13019 61 65 1:10,000 BG86 - 28 AS03 50 ug Neg Cntl 46 475 986 Pos Cntl 44473 39802 31871

(106) TABLE-US-00003 TABLE 2B Animal - Time after Plate Adjuvant & initial Sample ID Spike S1 Spike S2 Nucleocapsid Dilution Dosage vaccination 10 47883 0 96 1:10,000 BC43 - Day 42 Alum 50 ug 11 25917 0 0 1:100,000 BC43 - Day 42 Alum 50 ug 12 3597 14 0 .sup. 1:1,000,000 BC43 - Day 42 Alum 50 ug 13 45232 0 60 1:10,000 BC43 - Day 56 Alum 50 ug 14 16743 0 0 1:100,000 BC43 - Day 56 Alum 50 ug 15 2099 0 0 .sup. 1:1,000,000 BC43 - Day 56 Alum 50 ug 16 47513 156 7 1:10,000 BH56 - Day 42 AS03 50 ug 17 24243 31 21 1:100,000 BH56 - Day 42 AS03 50 ug 18 3976 11 0 .sup. 1:1,000,000 BH56 - Day 42 AS03 50 ug 19 45909 0 31 1:10,000 BH56 - Day 56 AS03 50 ug 20 18140 26 5 1:100,000 BH56 - Day 56 AS03 50 ug 21 1184 2 0 .sup. 1:1,000,000 BH56 - Day 56 AS03 50 ug 22 44095 53 0 1:10,000 BH95 - Day 42 Alum 50 ug 23 20940 26 0 1:100,000 BH95 - Day 42 Alum 50 ug 24 1845 2 28 .sup. 1:1,000,000 BH95 - Day 42 Alum 50 ug 25 45464 67 177 1:10,000 BH95 - Day 56 Alum 50 ug 26 17409 21 5 1:100,000 BH95 - Day 56 Alum 50 ug 27 1281 22 0 .sup. 1:1,000,000 BH95 - Day 56 Alum 50 ug 28 48319 71 0 1:10,000 BC11 - Day 42 AS03 50 ug 29 30010 10 0 1:100,000 BC11 - Day 42 AS03 50 ug 30 671 14 0 .sup. 1:1,000,000 BC11 - Day 42 AS03 50 ug 31 16796 4 0 1:10,000 BG86 - Single dose 8 AS03 50 ug weeks 32 1550 11 56 1:100,000 BG86 - Single dose 8 AS03 50 ug weeks 33 164 17 6 .sup. 1:1,000,000 BG86 - Single dose 8 AS03 50 ug weeks 34 0 64 15 1:1,000 BI37 - Day 0 Alum 25 ug 35 2 5 0 1:10,000 BI37 - Day 0 Alum 25 ug 36 8 6 25 1:100,000 BI37 - Day 0 Alum 25 ug 37 40318 112 0 1:1,000 BI37 - Day 14 Alum 25 ug 38 4456 35 28 1:10,000 BI37 - Day 14 Alum 25 ug 39 655 35 17 1:100,000 BI37 - Day 14 Alum 25 ug 40 49137 127 31 1:1,000 BI37 - Day 28 Alum 25 ug 41 32611 29 2 1:10,000 BI37 - Day 28 Alum 25 ug 42 5526 34 0 1:100,000 BI37 - Day 28 Alum 25 ug 43 0 56 15 1:1,000 BK78 - Day 0 AS03 25 ug 44 36 46 27 1:10,000 BK78 - Day 0 AS03 25 ug 45 45 34 0 1:100,000 BK78 - Day 0 AS03 25 ug 46 26462 49 87 1:1,000 BK78 - Day 14 AS03 25 ug 47 965 122 29 1:10,000 BK78 - Day 14 AS03 25 ug 48 92 28 37 1:100,000 BK78 - Day 14 AS03 25 ug 49 36563 67 31 1:1,000 BK78 - Day 28 AS03 25 ug 50 6352 34 17 1:10,000 BK78 - Day 28 AS03 25 ug 51 917 34 60 1:100,000 BK78 - Day 28 AS03 25 ug 52 1 0 160 1:1,000 BM52- Day 0 AS03 12.5 ug 53 15 64 60 1:10,000 BM52- Day 0 AS03 12.5 ug 54 12 79 49 1:100,000 BM52- Day 0 AS03 12.5 ug 55 17806 684 315 1:1,000 BM52- Day 14 AS03 12.5 ug 56 380 147 35 1:10,000 BM52- Day 14 AS03 12.5 ug 57 96 32 18 1:100,000 BM52- Day 14 AS03 12.5 ug 58 39459 440 29 1:1,000 BM52- Day 28 AS03 12.5 ug 59 4104 56 12 1:10,000 BM52- Day 28 AS03 12.5 ug 60 467 13 26 1:100,000 BM52- Day 28 AS03 12.5 ug 61 19 51 0 1:1,000 BC31 - Day 0 Alum 12.5 ug 62 0 0 2 1:10,000 BC31 - Day 0 Alum 12.5 ug 63 0 74 30 1:100,000 BC31 - Day 0 Alum 12.5 ug 64 9436 123 40 1:1,000 BC31 - Day 14 Alum 12.5 ug 65 426 27 0 1:10,000 BC31 - Day 14 Alum 12.5 ug 66 8 31 0 1:100,000 BC31 - Day 14 Alum 12.5 ug 67 36234 0 36 1:1,000 BC31 - Day 28 Alum 12.5 ug 68 7220 24 14 1:10,000 BC31 - Day 28 Alum 12.5 ug 69 160 9 8 1:100,000 BC31 - Day 28 Alum 12.5 ug Neg Cntl 221 1837 90 Neg Cntl 12 46 10 Pos Cntl 49197 44161 45329 Pos Cntl 47017 41224 46778

(107) TABLE-US-00004 TABLE 2C Days after Plate initial Adjuvant Sample ID Spike S1 Spike S2 Nucleocapsid Dilution Animal injection & Dose 1A 15544 333 104 1:10,000 BG86 84 AS03 50 ug - single dose 1B 1751 277 255 1:100,000 BG86 84 AS03 50 ug - single dose 1C 481 276 123 .sup. 1:1,000,000 BG86 84 AS03 50 ug - single dose 2A 35682 375 0 1:10,000 BH56 84 AS03 50 ug 2B 9450 313 259 1:100,000 BH56 84 AS03 50 ug 2C 1697 329 253 .sup. 1:1,000,000 BH56 84 AS03 50 ug 3A 39567 327 368 1:10,000 BC43 84 Alum 50 ug 3B 9891 95 124 1:100,000 BC43 84 Alum 50 ug 3C 1824 272 0 .sup. 1:1,000,000 BC43 84 Alum 50 ug

4A 37479 135 718 1:10,000 BH95 84 Alum 50 ug 4B 8790 161 143 1:100,000 BH95 84 Alum 50 ug 4C 1619 201 132 .sup. 1:1,000,000 BH95 84 Alum 50 ug 6A 8417 127 454 1:10,000 BC11 14 AS03 50 ug 6B 849 179 71 1:100,000 BC11 14 AS03 50 ug 6C 265 381 273 .sup. 1:1,000,000 BC11 14 AS03 50 ug 7A 13381 607 664 1:10,000 BC11 28 AS03 50 ug 7B 803 176 81 1:100,000 BC11 28 AS03 50 ug 7C 341 251 275 .sup. 1:1,000,000 BC11 28 AS03 50 ug 8A 45422 341 285 1:10,000 BC11 56 AS03 50 ug 8B 18249 173 165 1:100,000 BC11 56 AS03 50 ug 8C 3752 213 357 .sup. 1:1,000,000 BC11 56 AS03 50 ug 9A 42056 341 292 1:10,000 BC11 84 AS03 50 ug 9B 14271 316 163 1:100,000 BC11 84 AS03 50 ug 9C 2393 256 129 .sup. 1:1,000,000 BC11 84 AS03 50 ug 10A 44288 237 150 1:10,000 BI37 42 Alum 25 ug 10B 19432 189 119 1:100,000 BI37 42 Alum 25 ug 10C 4354 378 231 .sup. 1:1,000,000 BI37 42 Alum 25 ug 11A 43800 239 35 1:10,000 BI37 56 Alum 25 ug 11B 15355 138 75 1:100,000 BI37 56 Alum 25 ug 11C 2865 337 247 .sup. 1:1,000,000 BI37 56 Alum 25 ug 12A 38022 305 18 1:10,000 BI37 84 Alum 25 ug 12B 11812 294 86 1:100,000 BI37 84 Alum 25 ug 12C 1948 213 243 .sup. 1:1,000,000 BI37 84 Alum 25 ug 13A 46998 398 167 1:10,000 BK78 42 AS03 25 ug 13B 24787 267 116 1:100,000 BK78 42 AS03 25 ug 13C 6808 442 0 .sup. 1:1,000,000 BK78 42 AS03 25 ug 14A 45244 441 0 1:10,000 BK78 56 AS03 25 ug 14B 18294 209 84 1:100,000 BK78 56 AS03 25 ug 14C 3830 484 256 .sup. 1:1,000,000 BK78 56 AS03 25 ug 15A 41106 478 0 1:10,000 BK78 84 AS03 25 ug 15B 14414 275 83 1:100,000 BK78 84 AS03 25 ug 15C 3328 435 269 .sup. 1:1,000,000 BK78 84 AS03 25 ug 16A 46111 524 779 1:10,000 BM52 42 AS03 12.5 ug 16B 24586 113 300 1:100,000 BM52 42 AS03 12.5 ug 16C 7160 223 156 .sup. 1:1,000,000 BM52 42 AS03 12.5 ug 17A 41733 328 527 1:10,000 BM52 56 AS03 12.5 ug 17B 18140 284 157 1:100,000 BM52 56 AS03 12.5 ug 17C 3499 222 380 .sup. 1:1,000,000 BM52 56 AS03 12.5 ug 18A 36263 995 242 1:10,000 BM52 84 AS03 12.5 ug 18B 11912 525 277 1:100,000 BM52 84 AS03 12.5 ug 18C 1679 190 301 .sup. 1:1,000,000 BM52 84 AS03 12.5 ug 19A 43970 345 37 1:10,000 BC31 42 Alum 12.5 ug 19B 20155 255 25 1:100,000 BC31 42 Alum 12.5 ug 19C 3849 321 268 .sup. 1:1,000,000 BC31 42 Alum 12.5 ug 20A 40779 264 0 1:10,000 BC31 56 Alum 12.5 ug 20B 18417 298 196 1:100,000 BC31 56 Alum 12.5 ug 20C 2169 261 24 .sup. 1:1,000,000 BC31 56 Alum 12.5 ug 21A 32251 89 0 1:10,000 BC31 84 Alum 12.5 ug 21B 9977 314 197 1:100,000 BC31 84 Alum 12.5 ug 21C 1521 384 213 .sup. 1:1,000,000 BC31 84 Alum 12.5 ug 22A 43 249 129 1:10,000 BM21 0 AS03 12.5 ug 22B 267 215 186 1:100,000 BM21 0 AS03 12.5 ug 22C 225 270 141 .sup. 1:1,000,000 BM21 0 AS03 12.5 ug 23A 2694 331 96 1:10,000 BM21 14 AS03 12.5 ug 23B 621 202 190 1:100,000 BM21 14 AS03 12.5 ug 23C 297 339 225 .sup. 1:1,000,000 BM21 14 AS03 12.5 ug 24A 5672 361 75 1:10,000 BM21 28 AS03 12.5 ug 24B 905 39 168 1:100,000 BM21 28 AS03 12.5 ug 24C 91 129 139 .sup. 1:1,000,000 BM21 28 AS03 12.5 ug 25A 45613 555 118 1:10,000 BM21 49 AS03 12.5 ug 25B 23568 513 244 1:100,000 BM21 49 AS03 12.5 ug 25C 4134 293 248 .sup. 1:1,000,000 BM21 49 AS03 12.5 ug 26A 39959 429 19 1:10,000 BM21 63 AS03 12.5 ug 26B 15663 309 161 1:100,000 BM21 63 AS03 12.5 ug 26C 2263 240 273 .sup. 1:1,000,000 BM21 63 AS03 12.5 ug 27A 34745 371 47 1:10,000 BM21 91 AS03 12.5 ug 27B 9926 402 121 1:100,000 BM21 91 AS03 12.5 ug 27C 1197 299 380 .sup. 1:1,000,000 BM21 91 AS03 12.5 ug 28A 879 302 80 1:10,000 BD82 0 Alum 12.5 ug 28B 538 470 120 1:100,000 BD82 0 Alum 12.5 ug 28C 591 135 129 .sup. 1:1,000,000 BD82 0 Alum 12.5 ug 29A 3764 90 0 1:10,000 BD82 14 Alum 12.5 ug 29B 532 397 227 1:100,000 BD82 14 Alum 12.5 ug 29C 307 583 267 .sup. 1:1,000,000 BD82 14 Alum 12.5 ug 30A 9816 337 84 1:10,000 BD82 21 Alum 12.5 ug 30B 2407 371 169 1:100,000 BD82 21 Alum 12.5 ug 30C 285 462 245 .sup. 1:1,000,000 BD82 21 Alum 12.5 ug 31A 41086 344 128 1:10,000 BD82 35 Alum 12.5 ug 31B 19882 159 219 1:100,000 BD82 35 Alum 12.5 ug 31C 2912 333 73 .sup. 1:1,000,000 BD82 35 Alum 12.5 ug 32A 36345 245 57 1:10,000 BD82 49 Alum 12.5 ug 32B 13460 469 264 1:100,000 BD82 49 Alum 12.5 ug 32C 2585 149 9 .sup. 1:1,000,000 BD82 49 Alum 12.5 ug 33A 27977 400 185 1:10,000 BD82 77 Alum 12.5 ug 33B 7948 288 80 1:100,000 BD82 77 Alum 12.5 ug 33C 904 262 28 .sup. 1:1,000,000 BD82 77 Alum 12.5 ug 34A 62 274

150 1:10,000 BE83 0 Alum 25 ug 34B 385 232 217 1:100,000 BE83 0 Alum 25 ug 34C 142 306
292 .sup. 1:1,000,000 BE83 0 Alum 25 ug 35A 3232 69 94 1:10,000 BE83 14 Alum 25 ug 35B
804 343 34 1:100,000 BE83 14 Alum 25 ug 35C 129 222 161 .sup. 1:1,000,000 BE83 14 Alum
25 ug 36A 8300 150 58 1:10,000 BE83 21 Alum 25 ug 36B 1700 295 153 1:100,000 BE83 21
Alum 25 ug 36C 266 285 111 .sup. 1:1,000,000 BE83 21 Alum 25 ug 37A 38645 368 0
1:10,000 BE83 35 Alum 25 ug 37B 13433 320 151 1:100,000 BE83 35 Alum 25 ug 37C 3155 306
112 .sup. 1:1,000,000 BE83 35 Alum 25 ug 38A 35954 375 67 1:10,000 BE83 49 Alum 25 ug
38B 14846 190 89 1:100,000 BE83 49 Alum 25 ug 38C 3553 339 246 .sup. 1:1,000,000 BE83 49
Alum 25 ug 39A 23360 373 0 1:10,000 BE83 77 Alum 25 ug 39B 7836 72 31 1:100,000 BE83 77
Alum 25 ug 40A 12128 194 190 1:10,000 BG86 105 AS03 50 ug - single dose 40B 2292 161 159
1:100,000 BG86 105 AS03 50 ug - single dose 40C 261 187 191 .sup. 1:1,000,000 BG86 105
AS03 50 ug - single dose 41A 10969 161 0 1:10,000 BG86 139 AS03 50 ug - single dose 41B
1875 219 181 1:100,000 BG86 139 AS03 50 ug - single dose 41C 992 837 705 .sup. 1:1,000,000
BG86 139 AS03 50 ug - single dose 42A 176 57 120 1:10,000 BG86 MILK 1 AS03 50 ug - month
post single dose birth 42B 567 216 123 1:100,000 BG86 MILK 1 AS03 50 ug - month post single
dose birth 42C 217 283 153 .sup. 1:1,000,000 BG86 MILK 1 AS03 50 ug - month post single
dose birth 43A 9055 140 149 1:10,000 BP49 Birth Serum no vaccine Mother is BG86 43B 1410
121 235 1:100,000 BP49 Birth Serum no vaccine Mother is BG86 43C 388 311 152 .sup.
1:1,000,000 BP49 Birth Serum no vaccine Mother is BG86 44A 4525 35 0 1:10,000 BP49 4
weeks post no vaccine Mother is birth serum BG86 44B 1259 269 40 1:100,000 BP49 4 weeks post
no vaccine Mother is birth serum BG86 44C 286 261 223 .sup. 1:1,000,000 BP49 4 weeks post
no vaccine Mother is birth serum BG86 45A 23973 304 97 1:10,000 BC43 162 Alum 50 ug 45B
4896 272 161 1:100,000 BC43 162 Alum 50 ug 45C 649 212 208 .sup. 1:1,000,000 BC43 162
Alum 50 ug 46A 32326 192 36 1:10,000 BK78 142 AS03 25 ug 46B 11039 336 26 1:100,000
BK78 142 AS03 25 ug 46C 1694 418 226 .sup. 1:1,000,000 BK78 142 AS03 25 ug 47A 21598
206 0 1:10,000 BM72 120 AS03 25 ug 47B 5364 274 237 1:100,000 BM72 120 AS03 25 ug 47C
496 305 222 .sup. 1:1,000,000 BM72 120 AS03 25 ug 48A 18031 166 168 1:10,000 BH95 162
Alum 50 ug 48B 3372 304 173 1:100,000 BH95 162 Alum 50 ug 48C 562 350 191 .sup.
1:1,000,000 BH95 162 Alum 50 ug 49A 31501 102 147 1:10,000 BC11 150 AS03 50 ug 49B
8850 327 191 1:100,000 BC11 150 AS03 50 ug 49C 1716 447 257 .sup. 1:1,000,000 BC11 150
AS03 50 ug 50A 38099 414 277 1:10,000 BM52 142 AS03 12.5 ug 50B 16407 288 305
1:100,000 BM52 142 AS03 12.5 ug 50C 3171 304 136 .sup. 1:1,000,000 BM52 142 AS03 12.5
ug 51A 358 131 191 1:10,000 BJ55 0 None - Control 51B 443 363 357 1:100,000 BJ55 0 None -
Control 51C 138 371 117 .sup. 1:1,000,000 BJ55 0 None - Control 52A 19609 216 139 1:10,000
BI37 150 Alum 25 ug 52B 4058 261 214 1:100,000 BI37 150 Alum 25 ug 52C 465 148 32 .sup.
1:1,000,000 BI37 150 Alum 25 ug 53A 31360 392 175 1:10,000 BH56 169 AS03 50 ug 53B 8082
202 146 1:100,000 BH56 169 AS03 50 ug 53C 1013 342 97 .sup. 1:1,000,000 BH56 169 AS03
50 ug 54A 20996 282 110 1:10,000 BE83 118 Alum 25 ug 54B 4122 195 75 1:100,000 BE83 118
Alum 25 ug 54C 534 319 226 .sup. 1:1,000,000 BE83 118 Alum 25 ug 55A 433 353 151
1:10,000 BI98 0 None - Control 55B 290 186 114 1:100,000 BI98 0 None - Control 55C 129 297
190 .sup. 1:1,000,000 BI98 0 None - Control 56A 21633 245 129 1:10,000 BI21 56 Variant
COVID Vaccine AS03 25 ug 56B 5705 570 216 1:100,000 BI21 56 Variant COVID Vaccine AS03
25 ug 56C 840 228 155 .sup. 1:1,000,000 BI21 56 Variant COVID Vaccine AS03 25 ug 57A
10191 173 239 1:10,000 BC31 148 Alum 12.5 ug 57B 1927 340 170 1:100,000 BC31 148 Alum
12.5 ug 57C 217 480 353 .sup. 1:1,000,000 BC31 148 Alum 12.5 ug 58A 25739 262 179
1:10,000 BK44 56 Variant COVID Vaccine AS03 25 ug 58B 4927 278 91 1:100,000 BK44 56
Variant COVID Vaccine AS03 25 ug 58C 1183 183 113 .sup. 1:1,000,000 BK44 56 Variant
COVID Vaccine AS03 25 ug 59A 20341 177 4 1:10,000 BD82 118 Alum 12.5 ug 59B 3719 264
238 1:100,000 BD82 118 Alum 12.5 ug 59C 410 377 318 .sup. 1:1,000,000 BD82 118 Alum 12.5
ug 60A 38868 621 413 1:10,000 BM52 148 AS03 12.5 ug 60B 14327 303 250 1:100,000 BM52

148 AS03 12.5 ug 60C1649 214 132 .sup. 1:1,000,000 BM52 148 AS03 12.5 ug 61A 26273 155
0 1:10,000 BM21 125 AS03 12.5 ug 61B 5330 236 108 1:100,000 BM21 125 AS03 12.5 ug 61C
861 340 169 .sup. 1:1,000,000 BM21 125 AS03 12.5 ug 62A 19650 161 0 1:10,000 BI21 84
Variant COVID Vaccine AS03 25 ug 62B 5896 307 69 1:100,000 BI21 84 Variant COVID Vaccine
AS03 25 ug 62C 747 244 229 .sup. 1:1,000,000 BI21 84 Variant COVID Vaccine AS03 25 ug
63A 20018 179 39 1:10,000 BK44 84 Variant COVID Vaccine AS03 25 ug 63B 3860 184 53
1:100,000 BK44 84 A Variant COVID Vaccine S03 25 ug 63C 614 201 54 .sup. 1:1,000,000
BK44 84 Variant COVID Vaccine AS03 25 ug 66A 0 56 131 1:10,000 BK44 0 Variant COVID
Vaccine AS03 25 ug 66B 144 450 345 1:100,000 BK44 0 Variant COVID Vaccine AS03 25 ug 66C
224 411 146 .sup. 1:1,000,000 BK44 0 Variant COVID Vaccine AS03 25 ug 67A 1739 73 60
1:10,000 BK44 28 Variant COVID Vaccine AS03 25 ug 67B 415 306 202 1:100,000 BK44 28
Variant COVID Vaccine AS03 25 ug 67C 19 229 161 .sup. 1:1,000,000 BK44 28 Variant COVID
Vaccine AS03 25 ug 68A 25898 257 0 1:10,000 BK44 56 Variant COVID Vaccine AS03 25 ug
68B 7243 332 446 1:100,000 BK44 56 Variant COVID Vaccine AS03 25 ug 68C 1135 174 92
.sup. 1:1,000,000 BK44 56 Variant COVID Vaccine AS03 25 ug 69A 11 215 103 1:10,000 BI21
0 Variant COVID Vaccine AS03 25 ug 69B 95 0 50 1:100,000 BI21 0 Variant COVID Vaccine
AS03 25 ug 69C 152 165 102 .sup. 1:1,000,000 BI21 0 Variant COVID Vaccine AS03 25 ug 70A
2679 232 120 1:10,000 BI21 28 Variant COVID Vaccine AS03 25 ug 70B 429 335 2 1:100,000
BI21 28 Variant COVID Vaccine AS03 25 ug 70C 327 1562 839 .sup. 1:1,000,000 BI21 28
Variant COVID Vaccine AS03 25 ug 71A 22367 177 31 1:10,000 BI21 56 Variant COVID
Vaccine AS03 25 ug 71B 5939 281 116 1:100,000 BI21 56 Variant COVID Vaccine AS03 25 ug
71C 841 299 142 .sup. 1:1,000,000 BI21 56 Variant COVID Vaccine AS03 25 ug 72A 85 292 16
1:10,000 BM72 0 AS03 25 ug 72B 331 449 281 1:100,000 BM72 0 AS03 25 ug 72C 294 312 139
.sup. 1:1,000,000 BM72 0 AS03 25 ug 73A 4015 229 162 1:10,000 BM72 28 AS03 25 ug 73B
764 251 262 1:100,000 BM72 28 AS03 25 ug 73C 476 499 165 .sup. 1:1,000,000 BM72 28 AS03
25 ug 74A 35753 239 39 1:10,000 BM72 56 AS03 25 ug 74B 13297 277 123 1:100,000 BM72 56
AS03 25 ug 74C 2518 396 126 .sup. 1:1,000,000 BM72 56 AS03 25 ug

Example 3

(108) Further constructs containing certain SARS-CoV-2 immunogenic fragments were prepared and expressed in accordance with the methods disclosed herein. See, e.g., the first three paragraphs of the Example section. The protein expression of the constructs was subsequently measured after four days of expression. See Table 3; FIG. 6. All tested constructs expressed at a concentration of at least about 50 mg protein/L culture.

(109) TABLE-US-00005 TABLE 3 Protein yields for Protein A affinity-purified BA.X- monomer-Fc proteins after 4-days of expression. Strain Number Protein mg protein/L culture 2634 BA.X-1-monomer-Fc 85.9 (SEQ ID NO: 255) 2635 BA.X-2-monomer-Fc 58.3 (SEQ ID NO: 256) 2636 BA.X-3 -monomer-Fc 59.5 (SEQ ID NO: 257) 2637 BA.X-4-monomer-Fc 69.3 (SEQ ID NO: 258) 2638 BA.X-5 -monomer-Fc 47.9 (SEQ ID NO: 259) 2639 BA.X-6-monomer-Fc 51.4 (SEQ ID NO: 260) 2640 BA.X-7-monomer-Fc 73.4 (SEQ ID NO: 261) 2641 BA.X-8-monomer-Fc 91.0 (SEQ ID NO: 262) 2642 BA.X-9-monomer-Fc 48.2 (SEQ ID NO: 263) 2643 BA.X-10-monomer-Fc 75.7 (SEQ ID NO: 264)

(110) TABLE-US-00006 TABLE 4 Mutations within Monomer (with corresponding SEQ ID NO)
In Comparison to Wuhan wild-type variant Mutation BA.X-1 BA.X-2 BA.X-3 BA.X-4 BA.X-5
BA.X-6 BA.X-7 BA.X-8 BA.X-9 BA.X-10 Position (245) (246) (247) (248) (249) (250) (251)
(252) (253) (254) G339 G339D G339D G339D G339D G339D G339D G339D G339D G339D
G339D R346 R346K R346K S371 S371F S371F S371F S371L S373 S373P S373P S373P S373P
S373P S375 S375F S375F S375F T376 T376A T376A T376A T376A T376A T376A T376A
T376A D405 D405N D405N D405N D405N D405N D405N D405N D405N D405N R408 R408S
R408S R408S R408S R408S R408S R408S R408S R408S K417 K417N K417N K417N K417N
K417N K417N K417N K417N K417N N440 N440K N440K N440K N440K N440K

N440K N440K N440K N440K N440K G446 G446S G446S L452 L452R L452R L452R L452R
S477 S477N S477N S477N S477N S477N S477N S477N S477N S477N S477N T478 T478K
T478K T478K T478K T478K T478K T478K T478K T478K T478K E484 E484A E484A E484A
E484K E484K E484A E484A E484A E484K E484A Q493 Q493R Q493R Q493R Q493R Q493R
Q493R Q493R Q493R Q493R G496 G496S G496S Q498 Q498R Q498R Q498R Q498R Q498R
Q498R Q498R Q498R N501 N501Y N501Y N501Y N501Y N501Y N501Y N501Y N501Y
N501Y N501Y Y505 Y505H Y505H Y505H Y505H Y505H Y505H Y505H Y505H

(111) Table 4 (above) depicts the amino acid substitutions, and corresponding positions thereof, within immunogenic fragments BA.X-1-BA.X-10 (SEQ ID NOs: 245-254) in comparison to the Wuhan wild-type variant. Table 5 (below) depicts the amino acid substitutions, and corresponding positions thereof, within the listed natural SARS-CoV-2 variants. As revealed by comparison of Tables 4 and 5, the BA.X-1-BA.X-10 immunogenic fragments each contain a combination of mutations not present in any shown natural SARS-CoV-2 strain. The exemplary particular combination of mutations selected for each BA.X-1-BA.X-10 immunogenic fragment provides broad antigenic coverage over existing SARS-CoV-2 Spike protein variants. Without being bound to any theory, it is expected that the broad antigenic coverage should also provide full or partial protection against later-arising SARS-CoV-2 Spike protein variants as well. The constructs containing the BA.X-1-BA.X-10 immunogenic fragments are capable of being expressed at levels suitable for commercial vaccine production. See Table 3, FIG. 6.

(112) TABLE-US-00007 TABLE 5 Natural variant mutations in comparison to Wuhan wild-type variant Mutation Delta Omicron Omicron Position Alpha Beta Gamma Delta Plus Mu B.1.1.529
BA.2 G339 G339D G339D R346 R346K S371 S371L S371F S373 S373P S373P S375 S375F
S375F T376 T376A D405 D405N R408 R408S K417 K417N K417T K417N K417N K417N N440
N440K N440K G446 G446S L452 L452R L452R S477 S477N S477N T478 T478K T478K
T478K T478K E484 E484K E484K E484K E484A E484A Q493 Q493R Q493R G496 G496S
Q498 Q498R Q498R N501 N501Y N501Y N501Y N501Y N501Y N501Y Y505 Y505H Y505H

(113) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

(114) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A recombinant polypeptide comprising at least one immunogenic fragment of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein comprising an amino acid

- sequence with at least 93% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.
2. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least two SARS-CoV-2 spike glycoprotein immunogenic fragments.
 3. The recombinant polypeptide of claim 2, wherein two immunogenic fragments of the at least two immunogenic fragments comprise the same amino acid sequence.
 4. The recombinant polypeptide of claim 2, wherein two immunogenic fragments of the at least two immunogenic fragments comprise a different amino acid sequence.
 5. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 245.
 6. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 248.
 7. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 252.
 8. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of SEQ ID NO: 254.
 9. The recombinant polypeptide of claim 1, wherein the polypeptide comprises at least one immunogenic fragment comprising an amino acid sequence with at least 93% sequence identity to the amino acid sequence of an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254, wherein the at least one immunogenic fragment further comprises one or more of the following amino acid substitutions: G339D, R346K, S371F, S373P, S375F, T376A, D405N, R408S, K417N, N440K, G446S, L452R, S477N, T478K, E484A, E484K, Q493R, G496S, Q498R, N501Y, and Y505H.
 10. The recombinant polypeptide of claim 2, wherein the at least two immunogenic fragments each comprise an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.
 11. The recombinant polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 255-264.
 12. The recombinant polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence with at least 93% sequence identity to the amino acid sequence selected from the group consisting of SEQ ID NOs: 275-284.
 13. The recombinant polypeptide of claim 2, wherein the at least two immunogenic fragments are connected to each other via a linker, and wherein the linker is a polypeptide comprising an amino acid sequence of 1-35 residues, wherein each residue is independently serine or glycine.
 14. The recombinant polypeptide of claim 1, wherein the at least one immunogenic fragment of the SARS-CoV-2 spike glycoprotein is connected to an antibody Fc region via a linker, wherein the linker comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 65, SEQ ID NO: 67, SEQ ID NO: 69, SEQ ID NO: 193, SEQ ID NO: 195, and SEQ ID NO: 197.
 15. A pharmaceutical composition comprising the recombinant polypeptide of claim 1 and a pharmaceutically acceptable carrier.
 16. The pharmaceutical composition of claim 15, wherein the pharmaceutical composition comprises an adjuvant.
 17. The pharmaceutical composition of claim 15, wherein the pharmaceutical composition does not comprise an adjuvant.
 18. A pharmaceutical composition comprising the recombinant polypeptide of claim 2 and a

pharmaceutically acceptable carrier.

19. A pharmaceutical composition comprising a recombinant polypeptide, the recombinant polypeptide comprising at least two immunogenic fragments of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) spike glycoprotein, and wherein at least one of the at least two immunogenic fragments comprises an amino acid sequence with at least 97% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.

20. The pharmaceutical composition of claim 19, wherein at least two of the at least two immunogenic fragments comprises an amino acid sequence with at least 97% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 245-254.
